

Metacognitive uncertainty in Correct and Error trials across inter-stimulus-intervals

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This pre-registration consists of 3 parts; 1. **considerations of models and empirical data of perceptual decision making tasks with confidence ratings** 2. **Outlining two computational models** 3. **Design, methods, hypotheses and sample size considerations of the current study.**

These three parts are covered in 3 separate documents, with this document including them all.

Part 1: Perceptual Decision-Making with Confidence

Metacognition, the ability to reflect on one's own cognitive processes, is commonly studied through confidence ratings in decision-making tasks. In this document we highlight empirical findings and experimental considerations when investigating perceptual decision making with confidence ratings.

We believe that there are three main aspects of confidence ratings and their relationship to choices.

- 1) Confidence ratings reflect a noisy, subjective probability of being correct or self-consistency, that incorporates response biases rather than purely objective performance ([Caziot and Mamassian 2021](#); [Sánchez-Fuenzalida et al. 2025](#); [Mihali et al. 2023](#); [Navajas et al. 2017](#)).
- 2) Confidence judgments underperform relative to first-order accuracy, (i.e. 90% Accuracy might produce 80% confidence) ([Shekhar and Rahnev 2021](#)).
- 3) Metacognition and error-monitoring are functionally related systems. Being confident about an error is a form of error-monitoring ([Yeung and Summerfield 2012](#); [Stephen M. Fleming and Daw 2017](#); [Özcel and Balci 2024](#)).

Models of perceptual decision making and metacognition

Computational models of metacognition typically fall into two categories: **static models** based on Signal Detection Theory (SDT) and **dynamic models** based on evidence accumulation (EA).

Static models describe choice and confidence as originating from a static process of acquiring a fixed evidence per trial. Confidence ratings are then assumed to be computed based on this evidence alone or in conjunction with additional independent information.

Dynamic models additionally account for response times by assuming that evidence is accumulated over a time period. When a decision boundary is reached (enough evidence for a choice has been collected), a choice is made. Confidence ratings are then thought to be generated by additional evidence accumulation in the post-decisional time window.

Both of these approaches can be understood as (approximately) optimal solutions to partially observable Markov decision processes under different assumptions. In most perceptual decision making tasks these two types of models are not mutually exclusive and can be seen as extensions of one another.

Patterns of Confidence ratings

Regardless of originating from EA or SDT, a good metacognitive model needs to be able to capture several empirical patterns observed in the literature of both binary choices and confidence ratings.

1. **Performance increases with stimulus intensity:** higher stimulus intensity yields higher accuracy
2. **Confidence curves shifts in tandem with binary choice curves:** As the psychometric function, governing the choice probability, moves from unbiased (threshold at 0) so does the confidence curves. Meaning that the lowest confidence ratings for correct trials are at the stimulus value of the threshold (Caziot and Mamassian 2021; Sánchez-Fuenzalida et al. 2025; Mihali et al. 2023).
3. **Confidence increases with stimulus intensity:** higher stimulus intensities yield higher Confidence ratings (conditional on accuracy, see below) (Hangya et al. 2016; Sanders et al. 2016)
4. **Confidence varies with accuracy:** The relationship between confidence and stimulus intensity depends on response correctness, with correct trials showing higher confidence than incorrect. (Hangya et al. 2016; Sanders et al. 2016).
 - a. **“Folded-X” pattern:** Confidence sometimes increases with stimulus intensity for correct responses but decreases (or plateaus) for errors, considered to be a signature of metacognition (Hangya et al. 2016; Sanders et al. 2016).
 - b. **“Double-increase” pattern:** Confidence increases with stimulus intensity for both correct and incorrect responses

Recent work suggests these two differing patterns of confidence, could potentially arise from different experimental manipulations (Xue et al. 2026; Fung et al. 2025):

- **Folded-X:** Emerges when manipulating decision-relevant stimulus properties (e.g., coherence in RDM)
- **Double-increase:** Occurs when manipulating auxiliary difficulty markers (e.g., stimulus visibility).

These recent studies investigating 4a and 4b show these patterns using a confidence scale that ranges from guessing to certain. This highly limits the information the incorrect side of the confidence curve holds, as we will argue for in the next section. Furthermore, if subjects are not able express that they made an error, one might argue that this cannot be error-monitoring, even if confidence ratings for incorrect trials decrease as stimulus intensity increases.

Confidence patterns may reflect error-monitoring processes, but this can only be accessed when the confidence scale allows participants to explicitly indicate that they believe they made a mistake. Furthermore, as expressed below, confidence ratings need to be administered after the binary choice and not simultaneous with the choice in order for subjects to make a mistake and then be aware of it. If subjects choose and rate simultaneously a folded X-pattern would indicate that the participant is intentionally giving an incorrect answer.

To summarize; only a full confidence scale ranging from incorrect to correct and a sequential design of choice followed by confidence rating give subjects the ability to demonstrate error-monitoring which would lead to a full meaningful folded-X pattern. This folded-X pattern would have confidence ratings for incorrect trials go below the middle of the scale which expresses “guessing”. Only in such a case could the folded-X reflect functioning error-monitoring (confidence decreases for errors), while the double-increase pattern would suggest no error-monitoring. below we define and describe in more detail how experimental designs and measurement scale can influence the type of data and the interactions one might observe.

Confidence scales, stimulus granularity, and measurement considerations

Here we outline several methodological considerations, and argue that some of these will inevitably determine the types of empirical data one can observe.

Confidence measurement: Confidence is collected either discretely or continuously. Some models of metacognition (i.e. M-ratio) require discrete data, leading many experiments to collect or discretize continuous confidence ratings (Maniscalco and Lau 2012; Stephen M. Fleming 2017).

Stimulus intensity: Main stimulus axis that determine the difficulty of a trial. can be discrete or continuous.

Confidence scale construction: The particular anchors of a confidence scale.

- **Half-scale confidence.** A “half-scale” of confidence ranges from “guessing” to “confident”, this prevents expressing certainty about errors: agents cannot indicate they are sure they made a mistake. This generally produces “half-folded-X” patterns (flatish incorrect branch) rather than a full folded-X.
- **Full-scale confidence.** A “full scale” of confidence ranges from “confidently incorrect” to “confidently correct” enabling subjects to express error-monitoring and can therefore produce a full folded-X patterns.
- **Simultaneous choice–confidence.** When choice and confidence are produced together. With such an experimental paradigm a double-increase pattern should be expected, as otherwise an agent would have to deliberately choose the wrong answer.

These confidence patterns, which arise only under the aforementioned experimental constraints, primarily reshape how incorrect trials are distributed across stimulus intensity. This is due to the expressiveness of error-monitoring related to the scale, where only a full confidence scale allows participants to express that they made an error. A key question then becomes why agents make these errors in the first place, and especially how those errors can coexist with a sense of being wrong. Considering the above points, we believe that a task including continuous or many levels of, stimulus intensities and confidence ratings, is the most parsimonious and statistically powered way to investigate metacognition. The most important methodological consideration is a confidence scale that allows subjects to report that they made an error (i.e. a full-scale) is paramount.

Error-monitoring

Computational models of meta cognition have tried to explain error-monitoring in different ways. One of the main obstacles for these models is to answer the question of why would a rational agent make a mistake that they subsequently aware of.

One of the main ways this question has been addressed, across both modeling approaches, is to assume that there is additional information available to the rater compared to the binary choice, as if they were identical, genuine error-monitoring could not occur. (Stephen M. Fleming and Daw 2017).

Because of this Stephen M. Fleming and Daw (2017) formalized the second-order model. In this framework evidence for the action and evidence for the confidence rating is drawn from a multivariate normal with correlation ρ governing the amount of information sharing. If $\rho = 1$, the confidence judgement is based on the same information as the action (and no error-monitoring is possible, i.e. first-order model), however if $\rho = 0$ the confidence judgement is based on completely independent information.

A similar idea has been proposed in the dynamic models of perceptual decision making. In these models evidence for the initial choice arise from a noisy accumulation process until an evidence bound is hit and a choice is made. It is then further assumed that evidence keeps accumulating after the choice, in a post decision evidence accumulation window, which then gives rise to a confidence rating (Pleskac and Busemeyer

2010). This theory of post-decisional evidence accumulation (PDEA) of confidence ratings is one of the leading theoretical accounts of the generative process underlying both choice and confidence ratings that can explain error-monitoring (Desender et al. 2021). Below we provide a non-comprehensive review of studies investigating PDEA. Here we challenge the idea that additional independent information is necessary for the computation of confidence to the confidence patterns described above and as a results also error-monitoring. We end by arguing that the rater just needs access to the action provided by the actor in order to explain the empirical patterns.

Post Decisional Evidence Accumulation

There has been several approaches at determining how PDEA works. Generally it has been shown that PDEA increases metacognitive efficiency and has particular electro-physiological signatures (Desender et al. 2021).

PDEA unfolds within a remarkably short temporal window. Intracranial recordings in epilepsy patients performing mouse-tracking tasks reveal that post-decisional confidence processing occurs within 300 - 500 ms of choice initialization, with the pre-supplementary motor area associated with confidence and changes of mind (Goueytes et al. 2025; Resulaj et al. 2009). Neural recordings in rhesus monkeys has further demonstrated that anatomical regions implicated in choice formation are also the regions associated with confidence generation (Kiani and Shadlen 2009). This brief time course and overlapping neural circuitry suggests that PDEA, rather than reflecting slow deliberative processes, operates as a relatively automatic extension of the initial decision dynamics.

PDEA appears to be conditional on task demands and resource constraints. Under speed stress instructions (choice response times below 500 ms), confidence response times are much longer, and subjects show enhanced error monitoring, particularly for easy trial, compared to under accuracy stress. Under accuracy stress, when subjects have sufficient time for choice formation, choices are generally produced much slower and confidence judgments are made after a constant time, independent of difficulty and choice response time (Baranski and Petrusic 1998). Consistent with this pattern, subjects take significantly longer to respond when they know a confidence judgment is required compared to choice-only trials, suggesting that when not time-pressed, evidence accumulation for choice and confidence may occur pre-decisionally, i.e. in a parallel or single evidence-accumulation process (Petrusic and Baranski 2003; Baranski and Petrusic 2001).

Another way to investigate PDEA is to provide additional exposure to the stimulus after having committed to a choice, Stephen M. Fleming et al. (2018) showed that 300 ms of additional stimulus viewing after speed stressed choice (300ms) can increase confidence efficiency, with the strongest effect seen for a decrease in confidence ratings for incorrect trials.

Yu et al. (2015) found that manipulating inter-stimulus-intervals between choice and confidence produced changes in confidence ratings, here again particularly decreased confidence for errors, interestingly here it was found that additional exposure to the stimulus during this interval did not change the quality of the confidence ratings. This has been interpreted as internal evidence may be sufficient for post-decisional updating and additional exposure to the stimulus is not necessary.

In most of the highlighted studies a random dot motion stimulus is used as the perceptual decision making task. In such a task it has been shown that the performance of subjects asymptote with exposure times of only 300–600 ms, showing no improvement with longer presentations (Tsetsos et al. 2015). Crucially, this pattern holds even when stimulus-response mappings are manipulated, indicating that optimal decisions require only brief stimulus exposure, but if confidence ratings are required subjects tend to wait significantly longer to make their first response if not pressed for time.

Taken together, these findings suggest that when time permits, most confidence-relevant processing may occur pre-decisionally, with the most convincing evidence for PDEA being when subjects are time constrained in the primary decision. In such cases PDEA seem to increase metacognitive efficiency by mainly decreasing confidence ratings for incorrect choices.

An Alternative Account: Motor Control and Choice Variability

While the PDEA framework offers a mechanistically parsimonious explanation by using the same accumulation process for both choice and confidence, we propose a more general account that does not require additional independent information or extra PDEA. The short temporal window of PDEA (~300-500 ms) coincides with timescales consistent with motor control processes, meaning that we propose that what is characterized as PDEA may instead reflect motor awareness. This account can be further argued for given that under speed stress conditions subjects produce more errors, and these errors partly arise from reduced coordination between the central nervous system and effectors due to the time constraints. Motor control theory, particularly comparator models involving efference copy mechanisms, provides a framework for understanding these errors and how agents can become aware of them and subsequently correct them. When an agent initiates an action, an efference copy is generated and after commitment of the action it is compared against the expected sensory consequences, via a forward model. Any mismatch between predicted and actual motor output signals an error, and the magnitude of this prediction error is an indication of the degree of confidence in the choice. Subjects can therefore become aware that their executed action differs from their intended choice, enabling confidence ratings below guessing, without requiring PDEA about the stimulus itself.

From the above formulation we propose that error-monitoring can arise from choice variability, i.e. stochasticity in the mapping from evidence to choice. While most perceptual decision-making models assume deterministic choices given the acquired evidence, a stochastic choice process allows agents to produce errors that they can subsequently be aware of. Ordinarily, evidence variability and choice variability would be indistinguishable from binary responses alone, however, as will be shown below multivariate modeling of choices and confidence ratings allows for distinguishing these noise sources.

In the following section we derive one particular model incorporating choice variability, which can produce the empirical patterns of confidence and choices described earlier. This model can be expressed as a specific instantiation of Fleming and Daw’s second-order framework ([Stephen M. Fleming and Daw 2017](#)), but critically, without requiring independent information but still allowing for error-monitoring. The rater need only access the actor’s chosen action, not additional post-decisional evidence.

Part 2: Current Model

In this section we first describe the general structure of ideas of static models of perceptual decision making and confidence ratings. Next we describe, derive and simulate from our proposed model, demonstrating that it can produce some important patterns highlighted in the [Design considerations](#).

Static models of metacognition

In SDT, an actor makes a binary choice on each trial i , $a_i \in \{-1, +1\}$ based on some internal evidence e_i . The action reflects the actor’s inference on the true state of the stimulus $D_i \in \{-1, +1\}$.

In SDT, evidence is drawn from a normal distribution, as perception is a noisy process. The mean of this distribution reflects the true state and scales with stimulus intensity X_i .

$$e_i \sim \mathcal{N}(X_i D_i, \sigma_e)$$

The actor then produces an action by comparing the internal evidence to a criterion c :

$$a_i = \begin{cases} +1 & \text{if } e_i > c \\ -1 & \text{if } e_i \leq c \end{cases}$$

Within most frameworks, this criterion is thought to be static within an experimental setting for each agent.

Confidence Ratings Having made a binary choice, the agent is asked to give a confidence judgement in their choice. A particularly prominent way to describe this is assume that the agent computes the probability that the action was correct. Stephen M. Fleming and Daw (2017) proposed an approach within the framework discussed above that could also account for error-monitoring.

In their formulation confidence judgement are based on draws from a multivariate normal, which contains the evidence used for the action e_i and pseudo-independent information only accessible to the rater r_i . These two quantities are assumed to be correlated with ρ governing information sharing. I.e. if $\rho = 1$, the confidence judgement is based on the same information as the action, however if $\rho = 0$ the confidence judgement is based on completely independent information.

$$\begin{bmatrix} e_i \\ r_i \end{bmatrix} \sim \mathcal{N} \left(\begin{bmatrix} X_i D_i \\ X_i D_i \end{bmatrix}, \begin{bmatrix} \sigma_e^2 & \rho \cdot \sigma_e \cdot \sigma_m \\ \rho \cdot \sigma_e \cdot \sigma_m & \sigma_m^2 \end{bmatrix} \right)$$

The rater is then thought to use this information to compute the subjective probability of being correct which is used to generate a confidence rating. One of the main arguments for such model and other models with independent information was to explain error-monitoring i.e. agents' ability to know that they produced an error and give low (even below guessing) confidence ratings. The necessity of independent information for error monitoring follows from the fact that a rational agent who bases both choice and confidence on the same information (i.e. $\rho = 1$ in the above model), and whose choices are deterministic, cannot make an error while being aware of it. Any information indicating that the choice is incorrect would also determine the choice itself, thereby preventing the error

Efference copy (Choice Variability) model of metacognition (EA/CV - Meta)

Choices

Below we will argue that independent information is not required to explain error monitoring if we instead assume that agents show variability in their choices given the evidence. The difference from the above account is that instead of a deterministic decision policy (which is usually assumed in SDT), agents have a noisy decision process that can produce errors even though the evidence is in favor of the correct choice. Such errors could be related to speed stress conditions or more generally motor errors. By extending the standard SDT framework with the introduction of a stochastic choice policy it is possible for agents to display error monitoring. One way to implement this is to assume that the criterion varies from trial-by-trial: $c_i \sim \mathcal{N}(0, \sigma_c)$. This gives us:

$$a_i = \begin{cases} +1 & \text{if } e_i > c_i \\ -1 & \text{if } e_i \leq c_i \end{cases}$$

Due to the fact that both e_i and c_i are normally distributed, it is possible to marginalize their difference to obtain the probability of the action:

$$P(a = 1 \mid D_i, X_i) = \Phi \left(\frac{D_i \cdot X_i}{\sqrt{\sigma_e^2 + \sigma_c^2}} \right)$$

Before moving on to the confidence ratings we also assume that the evidence distribution might be biased for each subject. This assumption will help explain the fact that the psychometric function of the choices generally shift the other behavioral variables such as confidence ratings (Caziot and Mamassian 2021; Sánchez-Fuenzalida et al. 2025; Mihali et al. 2023; Navajas et al. 2017).

This leads to:

$$e_i \sim \mathcal{N}(X_i D_i - \beta, \sigma_e)$$

And when marginalized

$$P(a = 1 \mid D_i, X_i) = \Phi \left(\frac{D_i \cdot X_i - \beta}{\sqrt{\sigma_e^2 + \sigma_c^2}} \right) \quad (1)$$

From choices alone, only the sum $\sigma_e^2 + \sigma_c^2$ can be identified, as both noise sources equivalently flatten the psychometric function. However, by including confidence ratings these can be disentangled.

Confidence ratings

To compute confidence ratings, we assume that the agent internally computes the subjective probability that the choice was correct.

$$\text{confidence} = P(\text{correct} \mid r, a) = P(D = a \mid r)$$

Here we assume that the rater has access to a noisy readout of the evidence: $r_i \sim \mathcal{N}(e_i, \sigma_m)$, where σ_m is metacognitive noise.

This can be shown to yield: (See the full derivation in [Additional Derivations](#)).

$$P(\text{correct} \mid r_i, a, X_i) = \begin{cases} g\left(\frac{2 \cdot r_i \cdot X_i}{\sigma_e^2 + \sigma_m^2}\right) & \text{if } a = 1 \\ 1 - g\left(\frac{2 \cdot r_i \cdot X_i}{\sigma_e^2 + \sigma_m^2}\right) & \text{if } a = -1 \end{cases} \quad (2)$$

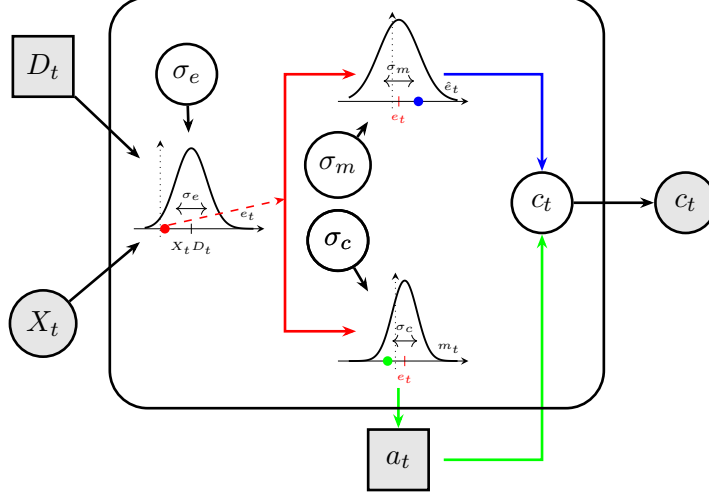
Given that we assume a uniform prior on D, i.e. $P(D_i = 1) = P(D_i = -1) = 0.5$, which represents the true state of the world (i.e. in the Random Dot Motion task whether the dots are moving up or down).

Where $g(\cdot)$ is the inverse logit transformation i.e. $g(x) = \frac{1}{1 + \exp(-x)}$.

From confidence alone, only the sum $\sigma_e^2 + \sigma_m^2$ can be identified. However, combining choices and confidence allows estimating all three parameters: σ_e , σ_c , and σ_m .

As can be seen from equation (2), to compute the probability of being correct the rater needs access to the stimulus intensity, X_i . Even though that it is assumed in some models ([Stephen M. Fleming et al. 2018](#)), we believe that the rater cannot have access to the objective stimulus intensity. We will get back to this and a final formulation without this term later. Interested readers are referred to ([Rausch and Zehetleitner 2019](#)), where simulations are conducted with different generative processes for the stimulus intensity.

Below we showcase a plate notation schematic of the above proposed model, demonstrating a visualization of the noise parameters.



The agent is presented with a stimulus that has a direction D_i (the true state of the world) and a strength X_i . An internal representation of the physical stimulus (evidence) is accessible to the agent e_i (red point), which is a noisy readout of the physical stimulus. This internal evidence representation is then sent to the pre-motor and motor cortex where an efferent signal is produced together with an efference copy (red split path). Due to additional noise (σ_c) in the processing and transmission of the signal to the limb generating the action, it is possible that the action generated, is in the opposite side of what the evidence entailed (green point). Simultaneously the efference copy might also be subjected to additional noise (σ_m), which is used together with the action to produce a prediction error in the comparator C_t . This prediction error is then used to compute confidence ratings C_t .

Next, we present simulations from the model to build intuition about the parameters and how they interact.

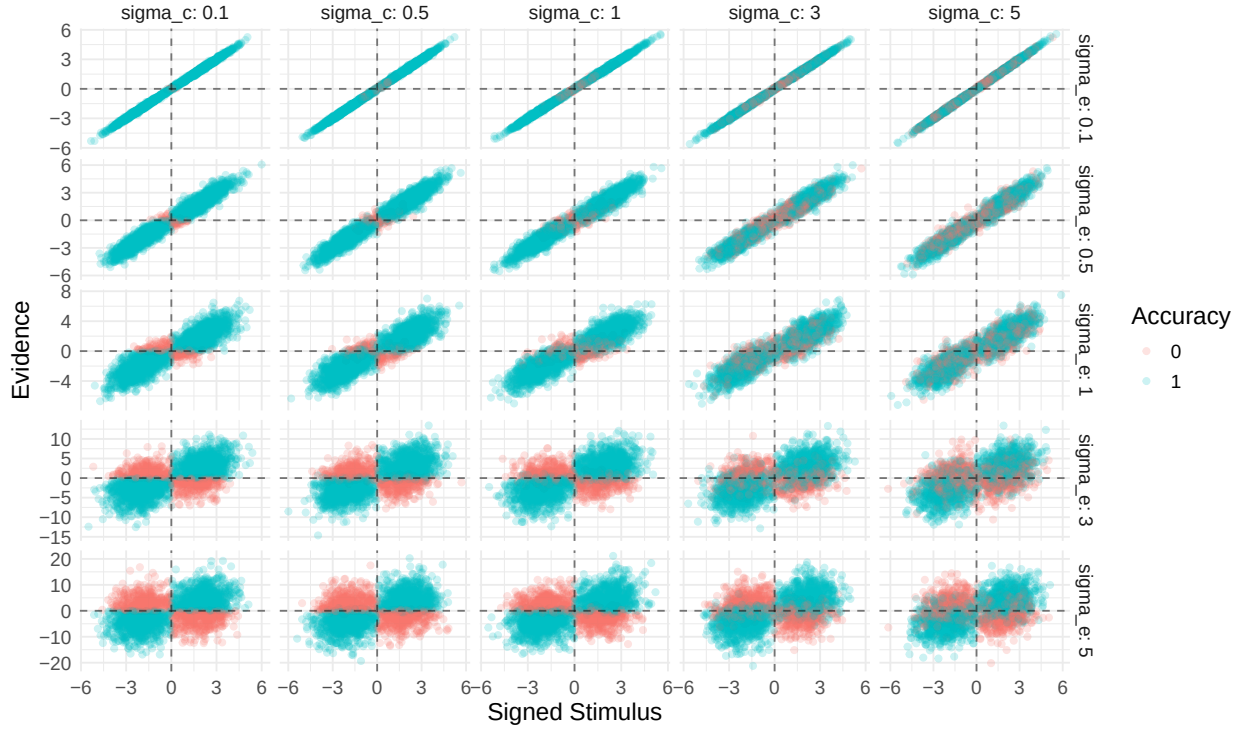
Simulations

Below we demonstrate that the model reproduces key behavioral patterns of choice and confidence ratings under different parameter combinations. Foreshadowing the results, our simulations show that the ratio $\frac{\sigma_e}{\sigma_c}$ determines the degree of error-monitoring and the main confidence patterns of folded-X or double increase.

- A low ratio (high σ_c low σ_e) produces folded-X (strong error-monitoring), while a high ratio (low σ_c high σ_e) produces double-increase (no error-monitoring).
- The metacognitive noise σ_m flattens the confidence curves regardless of accuracy.

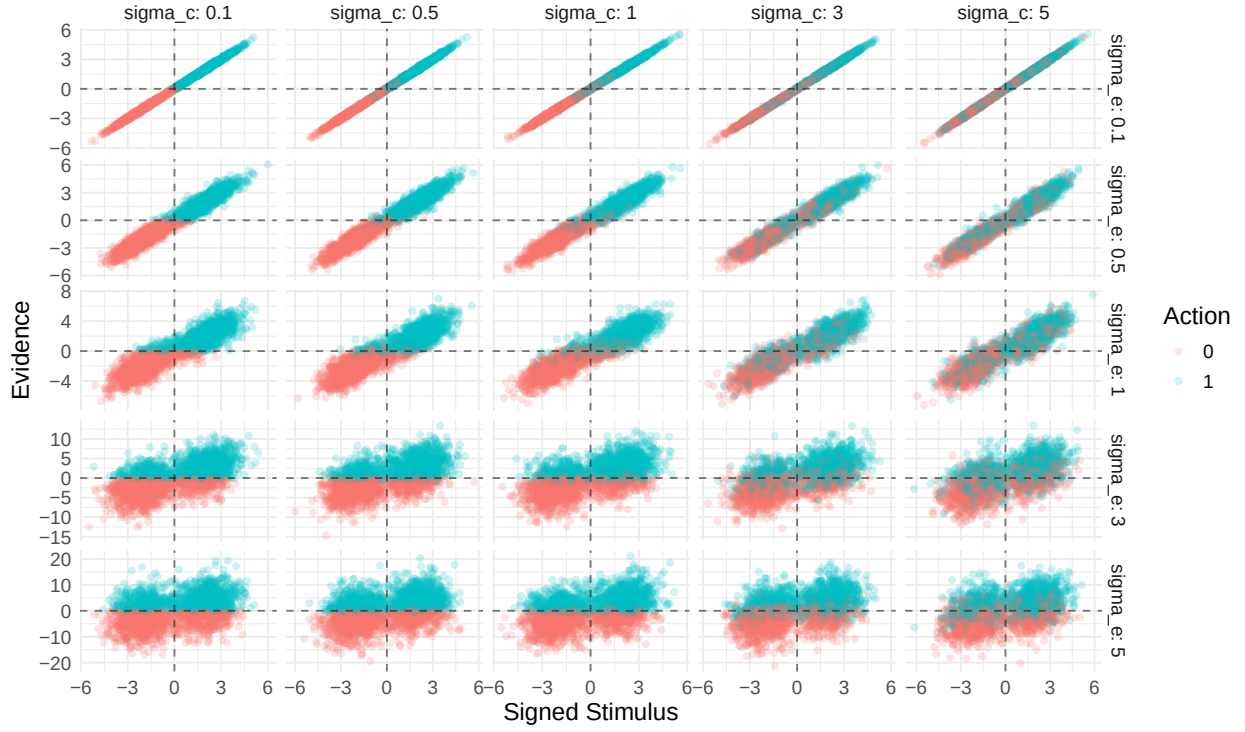
Note: For all simulations below we assume that the bias in evidence encoding β is 0. We will get back to this, which relates to the assumption of subjects have access to the objective stimulus intensity.

First, we might look at the distribution of internal evidence over the objective signed stimulus (i.e. the product of the stimulus intensity and the true state) across the two parameters governing the choices i.e. σ_e and σ_c being the evidence encoding and choice variability respectively.



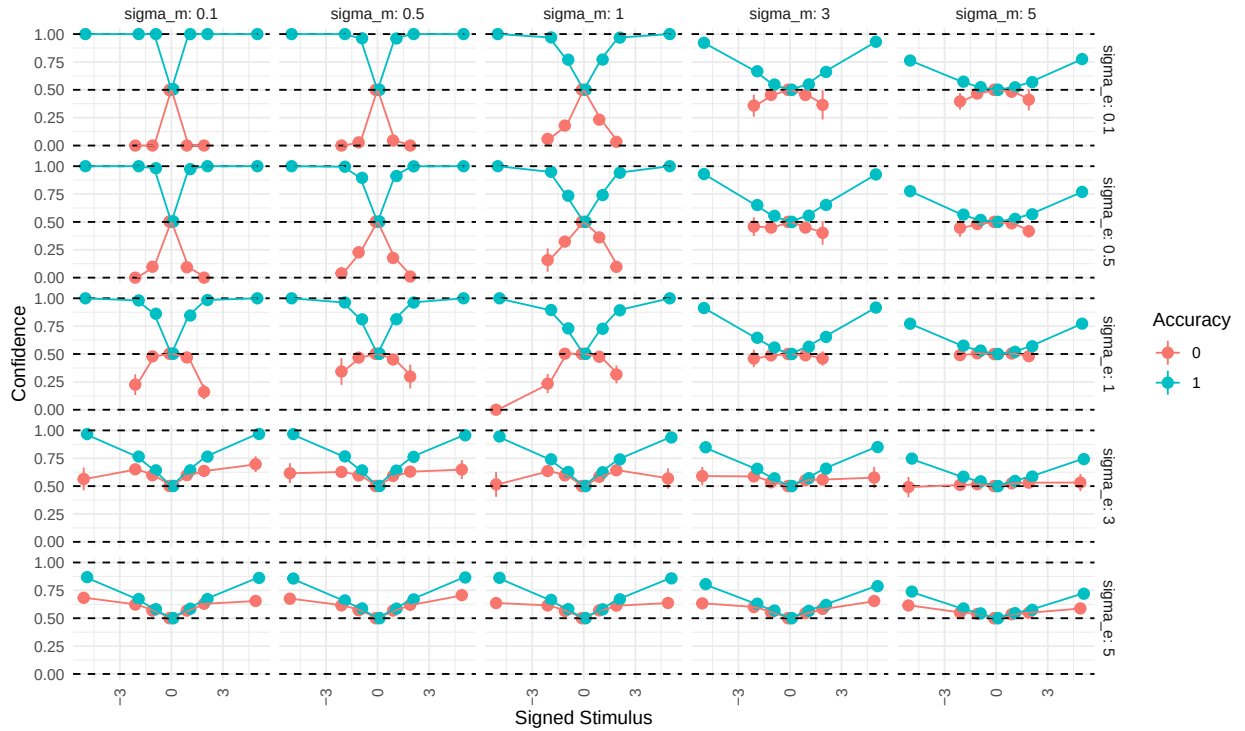
From the above plot we see that σ_e broadens the the range of evidence observed (essentially making it more noisy based on the same external stimulus), moving down the rows the y-axis greatly changes its range. When σ_c is low (left column) choices are consistent (when the the evidence is positive and the signed stimulus is positive (i.e. $D = 1$), a correct response is given). When σ_c increases this relationship essentially becomes more noisy, such that correct and incorrect choices are less distinguishable across the zero-evidence line and zero-signed-stimulus line.

Another way to display this is by plotting the actions instead of the accuracy:



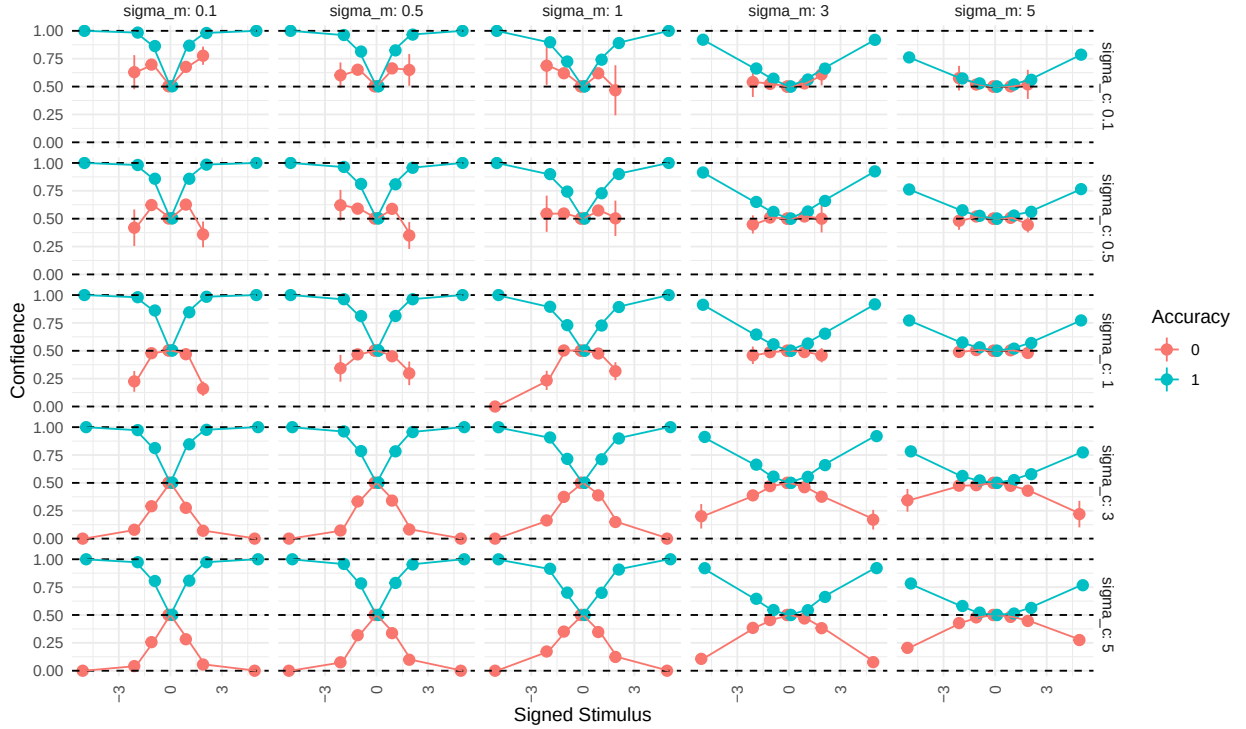
Next we look at the confidence ratings as a function of the three different sigmas.

First we fix $\sigma_c = 1$ and then we look at how σ_e & σ_m change confidence ratings as a function of the signed stimulus intensity:



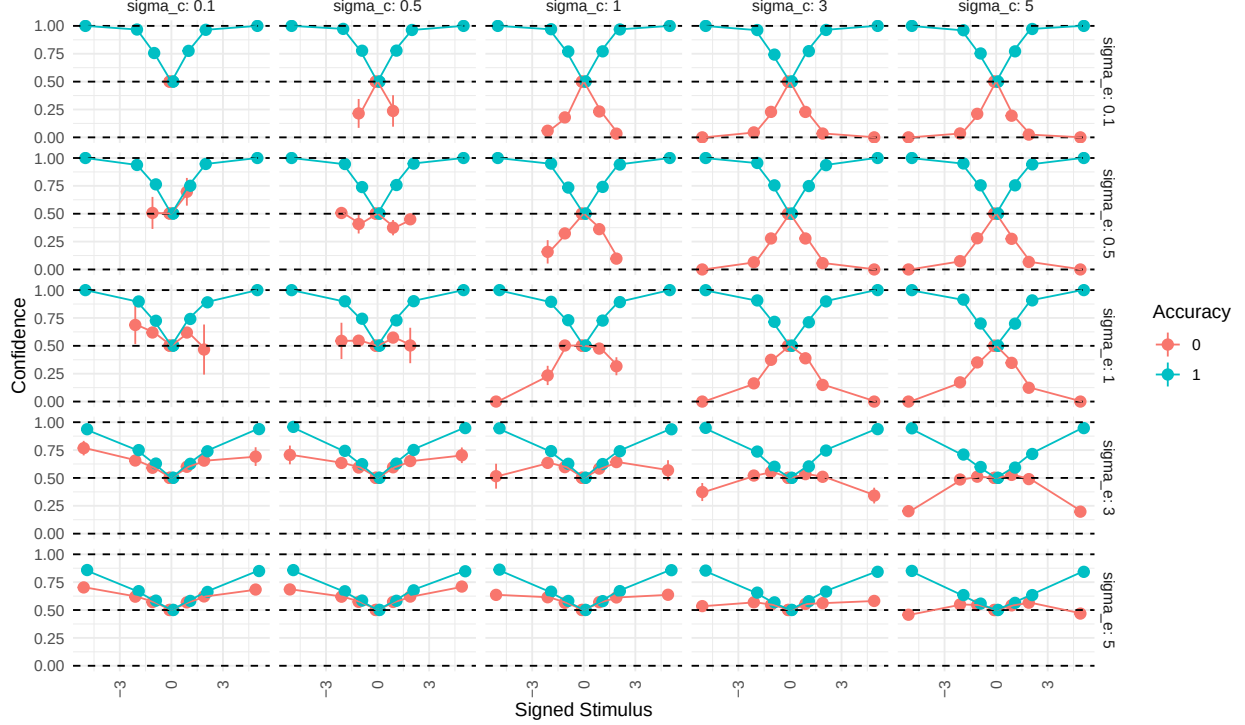
From the above plot we see that when we plot confidence on the Signed Stimulus scale we end up with a double increase pattern when σ_e is high (i.e. moving down the facets). Furthermore, when σ_e is low (moving up facets) we see a folded X pattern. Lastly, when meta uncertainty (σ_m) increase (moving left to right in the facets) both confidence curves becomes increasingly flat (i.e. closer to a flat line at 0.5) regardless of their correctness.

Next we fix $\sigma_e = 1$ and then we look at σ_c & σ_m :



Here we observe a similar patterns as before with $\sigma_e = 1$. Here a high σ_c reflects a folded X, whereas lower values (especially below 1) make a double increase. As before when meta uncertainty (σ_m) increase both confidence curves becomes increasingly flat (i.e. closer to a flat line at 0.5) regardless of their correctness.

lastly we plot the simulated data for varying σ_c and σ_e when $\sigma_m = 1$



Here we see that when $\sigma_e = \sigma_c$ (the diagonal), it becomes difficult to distinguish a folded-X pattern from a double-increase pattern, because incorrect trials cluster around 0.5 and show little dependence on stimulus intensity. Correct trials increase in confidence across all parameter combinations with stimulus intensity. Looking at the upper triangle (where $\sigma_c > \sigma_e$), we see folded-X patterns, whereas in the lower triangle ($\sigma_c < \sigma_e$) we see the opposite pattern, double increase patterns.

These simulations show that from the current model's perspective, the double increase and Folded-X pattern arise due to different ratios of uncertainty in the encoding and choice. When choice variability is high compared to stimulus encoding, a folded X pattern arises whereas choice variability is low compared to stimulus encoding a double increase pattern arises. Meta uncertainty (σ_m) is agnostic to the accuracy of the rater and when it increases it makes the confidence curves more flat regardless of correctness.

Fitting

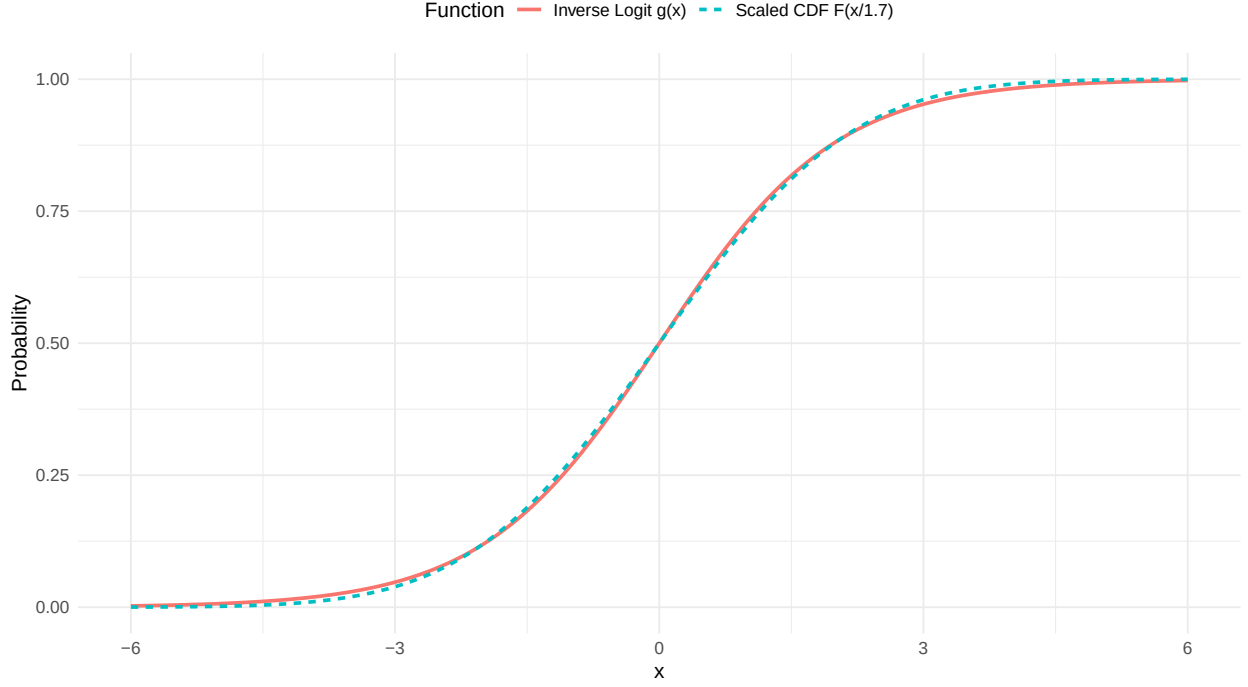
We want to fit this model using full Bayesian inference with Stan's NUTS sampler, treating the latent trial-by-trial evidence as parameters estimated jointly with the subject-by-subject noise parameters. Investigating our pilot data we saw that on a single-subject-level this model was possible to fit with sufficiently good sampler diagnostics. However, trying to fit this model in a hierarchical framework (i.e. drawing subject level parameters from group level parameters), we ran into sampling issues. As the current project is going to use a sequential sampling paradigm, a robust and efficient model is necessary. We therefore build a heuristic model that approximates the model shown above, without the need for estimation of trial-by-trial evidence.

The Heuristic model

While the above model is powerful, its computational complexity (especially in hierarchical settings) motivate a simpler heuristic approximation. The key challenge is that confidence requires trial-specific evidence r_i , which cannot be marginalized analytically, because the random variable is inside the inverse logit function.

The inverse logit function and the cumulative normal distribution are in practice close to being indistinguishable up to a constant i.e. $g(\cdot) \sim \Phi(\frac{1}{1.7})$, see plot below:

Comparison of Inverse Logit and Scaled Cumulative Normal



Using this approximation it is possible to marginalize out the trial-by-trial evidence and arrive at a computationally less expensive model (and an equation for the expectation of confidence) that uses the same parameters.

With this approximation we go from equation (2) to the following:

$$P(\text{correct} \mid r_i, a, X_i) = \begin{cases} \Phi\left(\frac{C \cdot r_i \cdot X_i}{\sigma_e^2 + \sigma_m^2}\right) & \text{if } a = 1 \\ 1 - \Phi\left(\frac{C \cdot r_i \cdot X_i}{\sigma_e^2 + \sigma_m^2}\right) & \text{if } a = -1 \end{cases} \quad (3)$$

Where $C = \frac{2}{1.7}$.

Focusing on the instance where the choice was one.

$$\Phi\left(\frac{C \cdot r_i \cdot X_i}{\sigma_e^2 + \sigma_m^2}\right)$$

We want to marginalize over r_i , i.e. calculate the expectation of above equation, across r_i .

This can be done using the general formula for the expectation of a random continuous variable.

$$\mathbb{E}(\mathbb{X}) = \int_{-\infty}^{\infty} x p(x) dx$$

Substituting the variables, with the following $b = \frac{C X_i}{\sigma_e^2 + \sigma_m^2}$

$$\int_{-\infty}^{\infty} \Phi(b \cdot r_i) p(r_i) dr_i$$

with $r_i \sim \mathcal{N}(\mu, \sigma)$

This integral can be shown to have the [solution](#):

$$\Phi\left(\frac{\mu \cdot b}{\sqrt{1 + \sigma^2 \cdot b^2}}\right)$$

Given that we have $r_i \sim \mathcal{N}(X_i D_i - \beta, \sqrt{\sigma_m^2 + \sigma_e^2})$, we get:

$$\Phi\left(\frac{\mathbb{E}(r_i) \cdot b}{\sqrt{1 + (\sigma_e^2 + \sigma_m^2) \cdot b^2}}\right)$$

Here we write the expectation of r_i , instead of the mean of the r_i distribution ($X_i D_i - \beta$), as this will be conditional on the choice (remember we are looking at $a_i = 1$).

Writing everything out and simplifying, we obtain:

$$\Phi\left(\frac{\mathbb{E}(r_i) \cdot C X_i}{\sigma_e^2 + \sigma_m^2} \cdot \frac{1}{\sqrt{1 + \frac{C^2 X_i^2}{\sigma_e^2 + \sigma_m^2}}}\right)$$

This equation gives the probability of being correct when the choice is 1. However, because the rater knows the actor's choice, and the rater's evidence is a noisy readout of the evidence used to generate that choice, observing the choice provides the rater with additional information about the underlying evidence. Mathematically this means that if the rater uses all the available evidence then the expectation of the available evidence needs to be conditional on the actors choice.

Equation (3), therefore now becomes:

$$P(\text{correct} \mid a, X_i) = \begin{cases} \Phi\left(\frac{\mathbb{E}(r_i|a=1) \cdot C X_i}{\sigma_e^2 + \sigma_m^2} \cdot \frac{1}{\sqrt{1 + \frac{C^2 X_i^2}{\sigma_e^2 + \sigma_m^2}}}\right) & \text{if } a = 1 \\ 1 - \Phi\left(\frac{\mathbb{E}(r_i|a=-1) \cdot C X_i}{\sigma_e^2 + \sigma_m^2} \cdot \frac{1}{\sqrt{1 + \frac{C^2 X_i^2}{\sigma_e^2 + \sigma_m^2}}}\right) & \text{if } a = -1 \end{cases} \quad (4)$$

It can be shown that this expectation given the choice is equal to the following equation: (For a full walk through of the derivation see below)

$$\mathbb{E}(r_i \mid a = 1) = (X_i D_i - \beta) + \rho \cdot \sqrt{\sigma_e^2 + \sigma_m^2} \cdot \lambda\left(\frac{X_i D_i - \beta}{\sqrt{\sigma_e^2 + \sigma_c^2}}\right)$$

For $a = -1$.

$$\mathbb{E}(r_i \mid a = -1) = (X_i D_i - \beta) - \rho \cdot \sqrt{\sigma_e^2 + \sigma_m^2} \cdot \lambda\left(-\frac{X_i D_i - \beta}{\sqrt{\sigma_e^2 + \sigma_c^2}}\right)$$

Here $\lambda(x) = \frac{\phi(x)}{\Phi(x)}$, and $\rho = \frac{\sigma_e^2}{\sqrt{(\sigma_e^2 + \sigma_c^2)} \cdot \sqrt{(\sigma_e^2 + \sigma_m^2)}}$

Final confidence model

Substituting these conditional expectations back into our marginalized confidence equation (4). This gives us after simplifying:

$$P(\text{correct} \mid a = 1, X_i, D_i) = \Phi \left(\left[\frac{(CX_i D_i - C\beta X_i)}{\sigma_e^2 + \sigma_m^2} + \rho \cdot \frac{\lambda \left(\frac{X_i D_i - \beta}{\sqrt{\sigma_e^2 + \sigma_c^2}} \right) CX_i}{\sqrt{\sigma_e^2 + \sigma_m^2}} \right] \cdot \frac{1}{\sqrt{1 + \frac{C^2 X_i^2}{\sigma_e^2 + \sigma_m^2}}} \right)$$

The above equation gives us a fully marginalized model for confidence ratings (for choices of 1) that depends only on the parameters $\Theta_1 = (\beta, \sigma_e, \sigma_c, \sigma_m)$ and the observed data (choices a_i , stimuli X_i and true states D_i), without requiring trial-by-trial estimation of latent evidence variables e_i and r_i .

Before interpreting the equation we need to revert back to one of the key assumptions of this model as was outline just after introducing equation (2). When the subject computes the subjective probability of being correct it is assumed that the subject has access to the stimulus intensity. Below we show which of the stimulus intensities this is related to (the two X_i terms that we highlight):

$$P(\text{correct} \mid a = 1, X_i, D_i) = \Phi \left(\frac{\left[(X_i D_i - \beta) + \rho \cdot \sqrt{\sigma_e^2 + \sigma_m^2} \cdot \lambda \left(\frac{X_i D_i - \beta}{\sqrt{\sigma_e^2 + \sigma_c^2}} \right) \right] \cdot C \mathbf{X}_i}{\sigma_e^2 + \sigma_m^2} \cdot \frac{1}{\sqrt{1 + \frac{C^2 \mathbf{X}_i^2}{\sigma_e^2 + \sigma_m^2}}} \right)$$

Interestingly these two stimulus intensities (shown in bold), especially the first one (which is not from the marginalization) serves to make sure that at intensity 0 the predicted confidence will always be 0.5. So not only is it a strong assumption that the agent has access to the objective stimulus strength, this assumption does not even hold given the empirical observation that the lowest confidence rating shifts with the bias which we highlighted in part 1. If confidence ratings are treated as noisy, biased subjective probabilities of being correct, the formulation above becomes problematic due to its reliance on an objective anchor at $X_i = 0$. Due to these consideration, supported by some empirical validation on our pilot data below, we believe that this particular stimulus intensity, coming from an unjustified assumption, needs to be accessed.

One interesting idea of how to deal with this, is to assume that the agent has a noisy readout of the stimulus strength, $\hat{X} \sim p(\hat{x}|\theta, c)$. The problem with such formulation is defining the probability distribution of the internal stimulus intensity. This probability distribution needs to be positively bound as $\hat{X} > 0$, which introduces an arbitrary choice, but more importantly makes it non trivial to marginalize the trial-by-trial readout. Another approach would be to replace this stimulus intensity value with a constant, which could vary across subjects. Such parameterization would serve as a placeholder for idiosyncratic differences in how subjects might use the stimulus intensity above and beyond what they should have access to.

Instead of opting for these options, we chose to assume that the agents do not have access to this stimulus intensity and therefore fix it to 1. This approach in its simplicity circumvents some of the troublesome computational aspects, as well as obeying to what the agents is assumed to have access to. This also importantly this makes it possible for the model to accommodate the empirical observation that the confidence curve follows the psychometric function of the binary choices. Below we will showcase that from our pilot data the two models (fixing this particular stimulus intensity to 1, or letting it vary based on the true stimulus intensity) have equivalent approximate leave one out cross-validation performance.

This means that the final equations become

$$P(\text{correct} \mid a = 1, X_i, D_i) = \Phi \left(\left[\frac{(CX_i D_i - C\beta)}{\sigma_e^2 + \sigma_m^2} + \rho \cdot \frac{\lambda \left(\frac{X_i D_i - \beta}{\sqrt{\sigma_e^2 + \sigma_c^2}} \right) C}{\sqrt{\sigma_e^2 + \sigma_m^2}} \right] \cdot \frac{1}{\sqrt{1 + \frac{C^2}{\sigma_e^2 + \sigma_m^2}}} \right) \quad (7a)$$

which for $a = -1$ becomes:

$$P(\text{correct} \mid a = -1, X_i, D_i) = 1 - \Phi \left(\left[\frac{(CX_i D_i - C\beta)}{\sigma_e^2 + \sigma_m^2} - \rho \cdot \frac{\lambda \left(-\frac{X_i D_i - \beta}{\sqrt{\sigma_e^2 + \sigma_c^2}} \right) C}{\sqrt{\sigma_e^2 + \sigma_m^2}} \right] \cdot \frac{1}{\sqrt{1 + \frac{C^2}{\sigma_e^2 + \sigma_m^2}}} \right) \quad (7b)$$

with:

$$\lambda(x) = \frac{\phi(x)}{\Phi(x)}$$

$$C = \frac{2}{1.7}$$

In the above formula it is noteworthy that the terms inside the cumulative normal distribution function have clear interpretations.

Firstly the common factor, is a correction factor for marginalization. thus we define:

$$K_i = \frac{1}{\sqrt{1 + \frac{C^2}{\sigma_e^2 + \sigma_m^2}}}$$

The two other terms can be thought to have more cognitive interpretations. Looking back at equation 3 we had (now with $X_i = 1$)

$$P(\text{correct} \mid r_i, a, X_i) = \begin{cases} \Phi \left(\frac{C \cdot r_i}{\sigma_e^2 + \sigma_m^2} \right) & \text{if } a = 1 \\ 1 - \Phi \left(\frac{C \cdot r_i}{\sigma_e^2 + \sigma_m^2} \right) & \text{if } a = -1 \end{cases} \quad (3)$$

One can show that the first term from equation (7a / 7b) is just the expectation of the argument to the cumulative normal distribution function in equation (3). This argument, from the derivation is the log-likelihood ratio, which can be interpreted as the average strength of evidence in favor of the true state. We might call this is I;

$$I_i = \mathbb{E} \left(\frac{C \cdot r_i}{\sigma_e^2 + \sigma_m^2} \right) = \frac{(CX_i D_i - C\beta)}{\sigma_e^2 + \sigma_m^2}$$

The latter term in equation (7a / 7b) adjusts this average evidence, because of the information contained within the action:

$$S_i = -\rho \cdot \frac{\lambda \left(-\frac{X_i D_i - \beta}{\sqrt{\sigma_e^2 + \sigma_c^2}} \right) C}{\sqrt{\sigma_e^2 + \sigma_m^2}}$$

This term serves the purpose of changing the mean expectation of the confidence to incorporate the information of the action. The addition of actions into the confidence computation have been proposed in the second order rater model from Fleming and Daw, where they also re-analyzed data to show this pattern empirically ([Stephen M. Fleming and Daw 2017](#)).

To summarize, confidence computation can in this model be thought of as a weighting of the expected normalized evidence with a correction factor for having committed to a choice. Below we introduce a couple of additional parameters in order to encompass idiosyncrasies in the task and participants, as well as probabilistic relate the above formulation to the observed confidence ratings.

Additional parameters We introduce 5 additional parameters. $\Theta_2 = (\kappa_c, \gamma, \mu_\beta, c_0, c_1)$.

- γ is the lapse rate on the binary choices this is a stimulus independent factor that shifts the asymptote of the psychometric function away from certainty i.e. the asymptotic becomes γ and $1 - \gamma$.
- μ_β is a meta cognitive bias parameter, serving as an independent vertical shift of confidence ratings.
- κ_c is the precision of the ordered beta likelihood.
- Lastly c_0, c_1 reflects the two cutpoints for the ordered beta likelihood. These two parameters help generalize the beta-distribution to outcomes that contain 0 and 1 ratings.

See (Kubinec 2023) for the ordered beta likelihood.

The complete set of parameters for the model are:

$$\Theta_s = \begin{bmatrix} \beta_s \\ \sigma_{e_s} \\ \sigma_{c_s} \\ \sigma_{m_s} \\ \mu_{\beta_s} \\ \kappa_{c_s} \\ \gamma_s \end{bmatrix} \sim \mathcal{N}(\mu_\Theta, \Sigma_\Theta)$$

$$\begin{aligned} \text{where } \beta_s &\in [-1, 1], \sigma_{e_s} > 0, \sigma_{c_s} > 0, \sigma_{m_s} > 0, \\ \mu_{\beta_s} &\in \mathbb{R}, \kappa_{c_s} > 0, \gamma_s \in [0, 0.5] \\ (c_{1_s}, c_{0_s}) &\sim \delta([1, 10, 1]) \end{aligned}$$

In the above formulation that $\mathcal{N}$ is a multivariate normal distribution with mean and covariance. Further all constraints on the parameters as written were accomplished by transformation, all positively constrained parameters were exponentiated, parameters with an upper and lower bound were transformed with the inverse logit function and multiplied appropriately. The constraint of the cut-points were applied by the prior of a Dirichlet distribution ($\delta()$). Note, the cutpoints are sampled on a subject-by-subject basis.

The full model becomes.

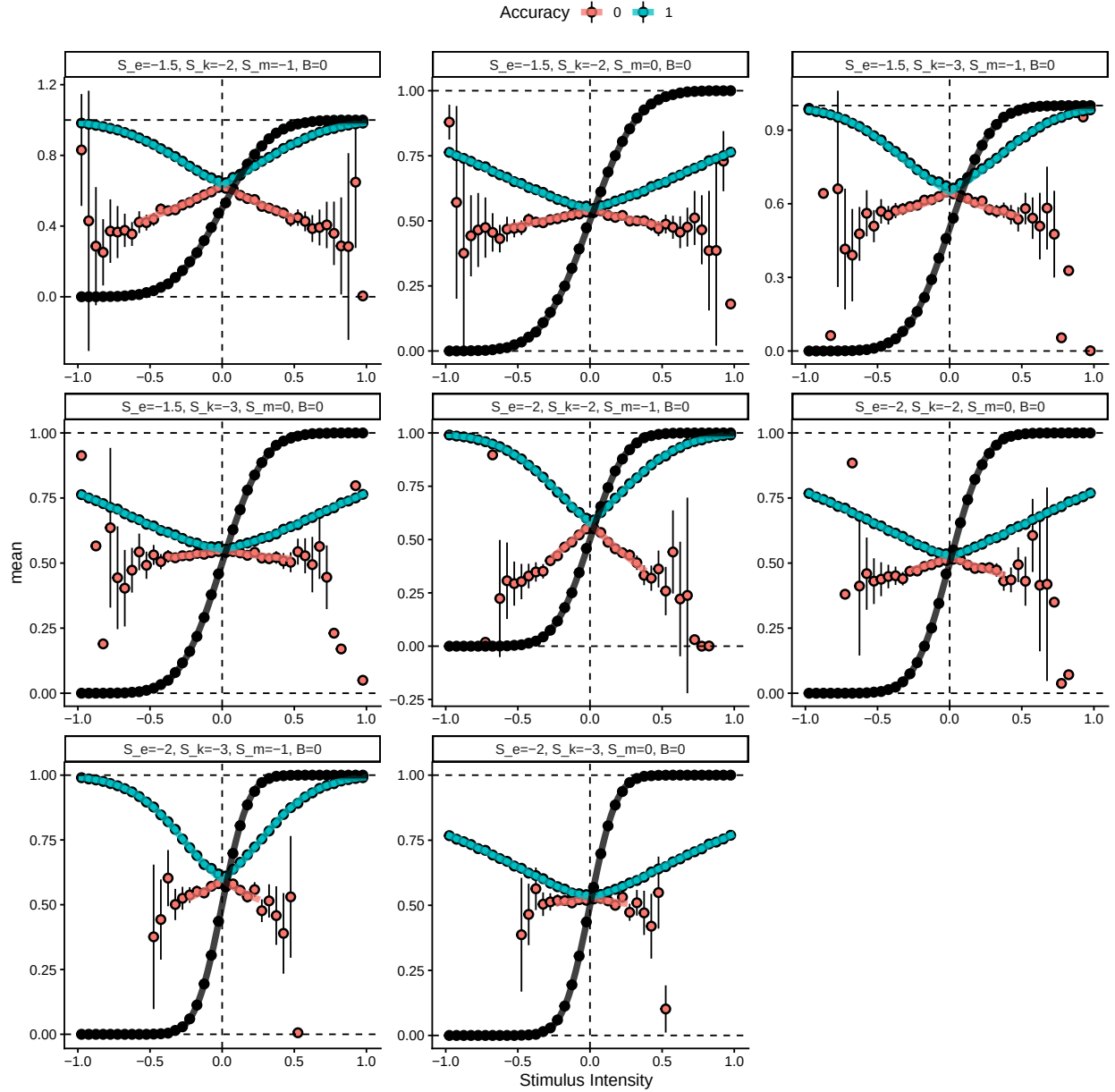
$$\begin{aligned} P(a = 1) &= \Phi \left(\frac{D_i \cdot X_i - \beta_s}{\sqrt{\sigma_{e_s}^2 + \sigma_{c_s}^2}} \right) \\ \mu_{c_i} &= \begin{cases} \Phi((I_i - S_i) \cdot C_i) & \text{if } a_i = 1 \\ 1 - \Phi((I_i + S_i) \cdot C_i) & \text{if } a_i = -1 \end{cases} \\ a_i &\sim \text{bernoulli}(p(a_i = 1)) \\ Conf_i &\sim \text{ord}\beta(\mu_{c_i}, \kappa_{c_s}, c_{0_s}, c_{1_s}) \end{aligned}$$

with

$$I_i = \frac{(CX_i D_i - C\beta)}{\sigma_{e_s}^2 + \sigma_{m_s}^2}, \quad S_i = -\rho_s \cdot \frac{\lambda \left(-\frac{X_i D_i - \beta_s}{\sqrt{\sigma_{e_s}^2 + \sigma_{c_s}^2}} \right) C}{\sqrt{\sigma_{e_s}^2 + \sigma_{m_s}^2}}, \quad K_i = \frac{1}{\sqrt{1 + \frac{C^2}{\sigma_{e_s}^2 + \sigma_{m_s}^2}}}$$

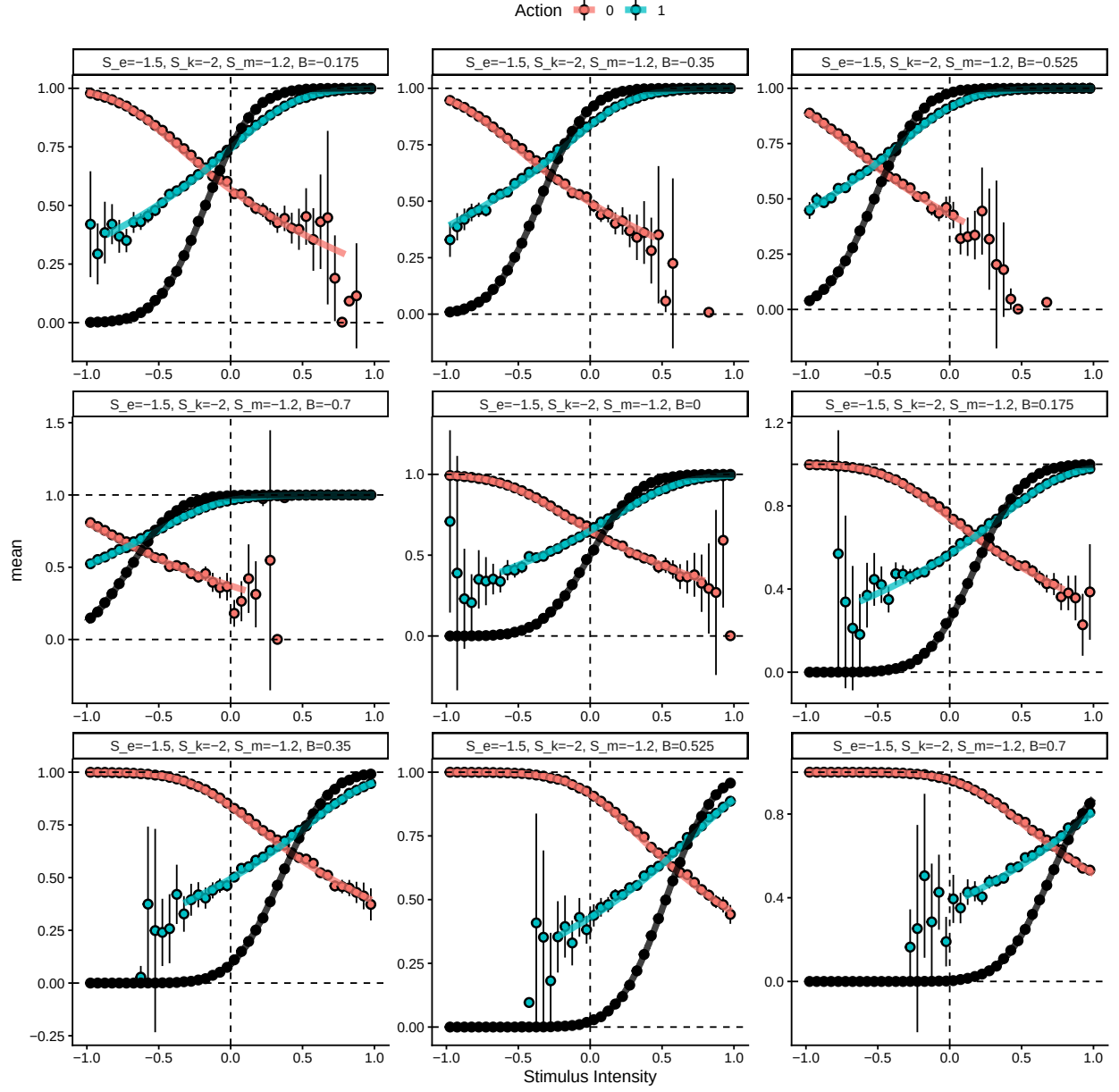
Next we show verification of the above formulated marginalized heuristic model, by simulating data from from equation (2) (with $X_i = 1$) averaging across the signed stimulus axis and displaying for the same parameter estimates (i.e. Θ_1) of the marginalized heuristic model.

The In the first validation step we simulate from different combinations of noise parameters (facets).



In the above plot we see the results from averring 5×10^5 simulated trials from equation (2), for each bin of the signed stimulus intensity scale (dots are mean and approximate 95 Interval). The thick solid lines represents the marginalized heuristic model predictions.

Next we simulate using fixed noise parameters and change the bias β . Note, that now we are displaying the action and not the accuracy of the choice.



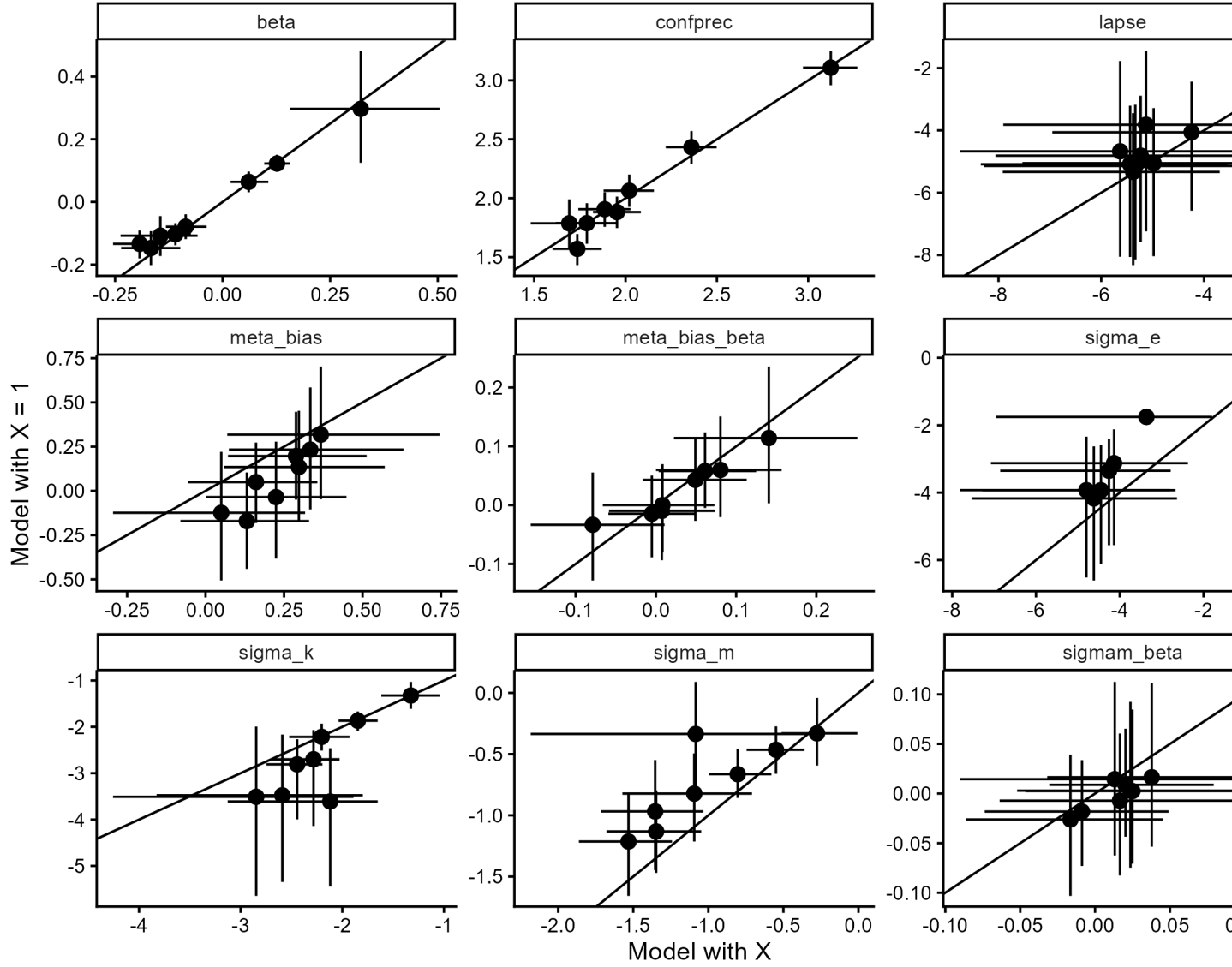
In the above plot we see the results from averaging 5×10^5 simulated trials from equation (2), for each bin of the signed stimulus intensity scale (dots are mean and approximate 95 Interval). The thick solid lines represents the marginalized heuristic model predictions.

Next, we showcase model comparison on approximate leave one trial out cross validation for the models assuming that the rater has access to the objective stimulus intensity and the reduced model with $X_i = 1$

Model comparison

As a sanity check on the assumption of $X_i = 1$, we fitted both hierarchical models to our pilot data and computed approximate leave one out cross-validation, to test if the assumption could be justified. This comparison showed that the two models were virtually indistinguishable with the difference in expected log probability density being -1.58 ± 9.9 , where generally it is thought that a difference below -4 and or more

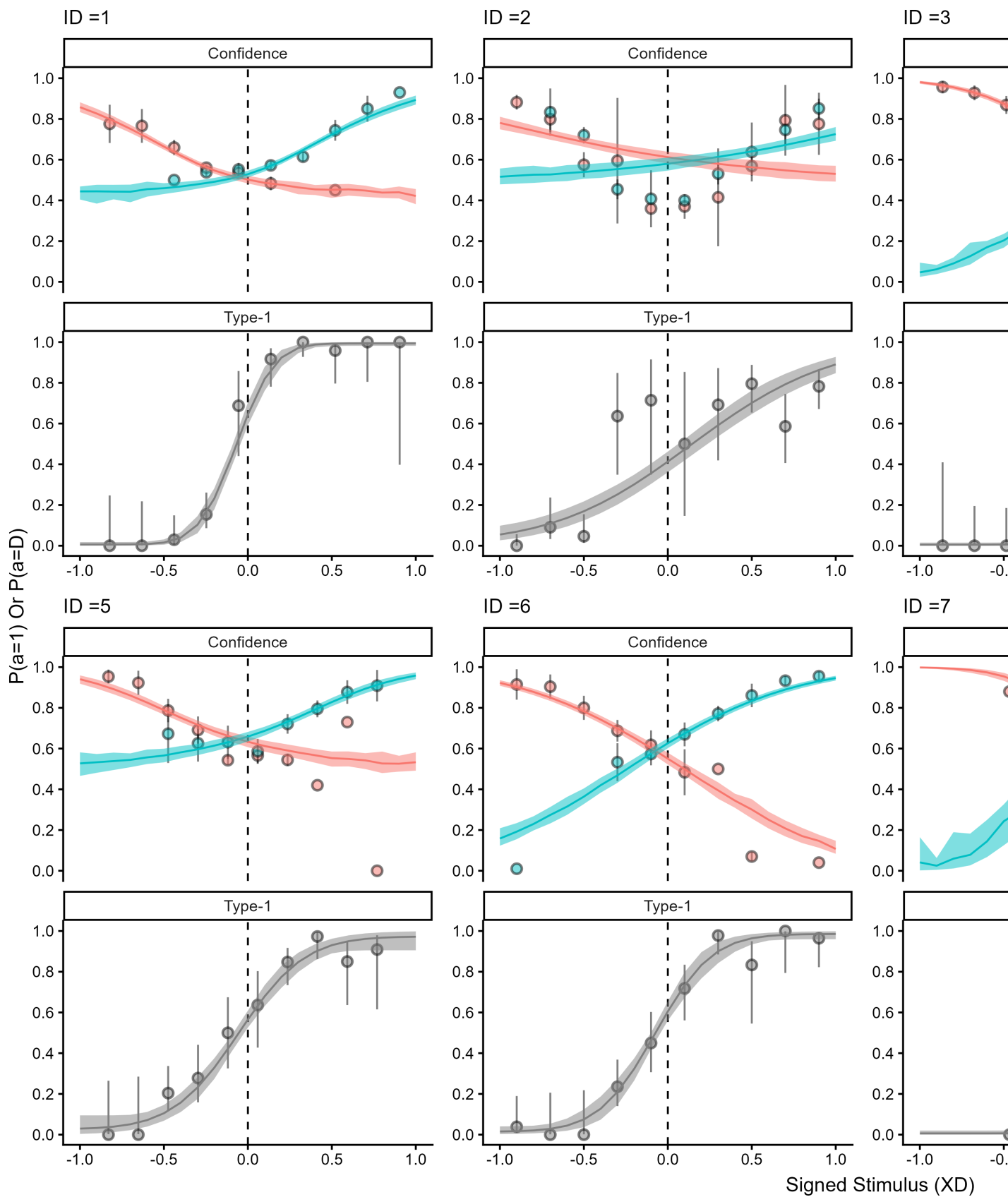
than two standard errors can be considered as evidence for one model over the other (REF), thus at least for our pilot data for predictive performance there seems to be no difference. Digging further into the distinction between these two models we below plot all pairwise parameters together with the identity line.



Generally the two models align well in their subject level parameter estimates, further justifying this simplification which makes theoretical sense.

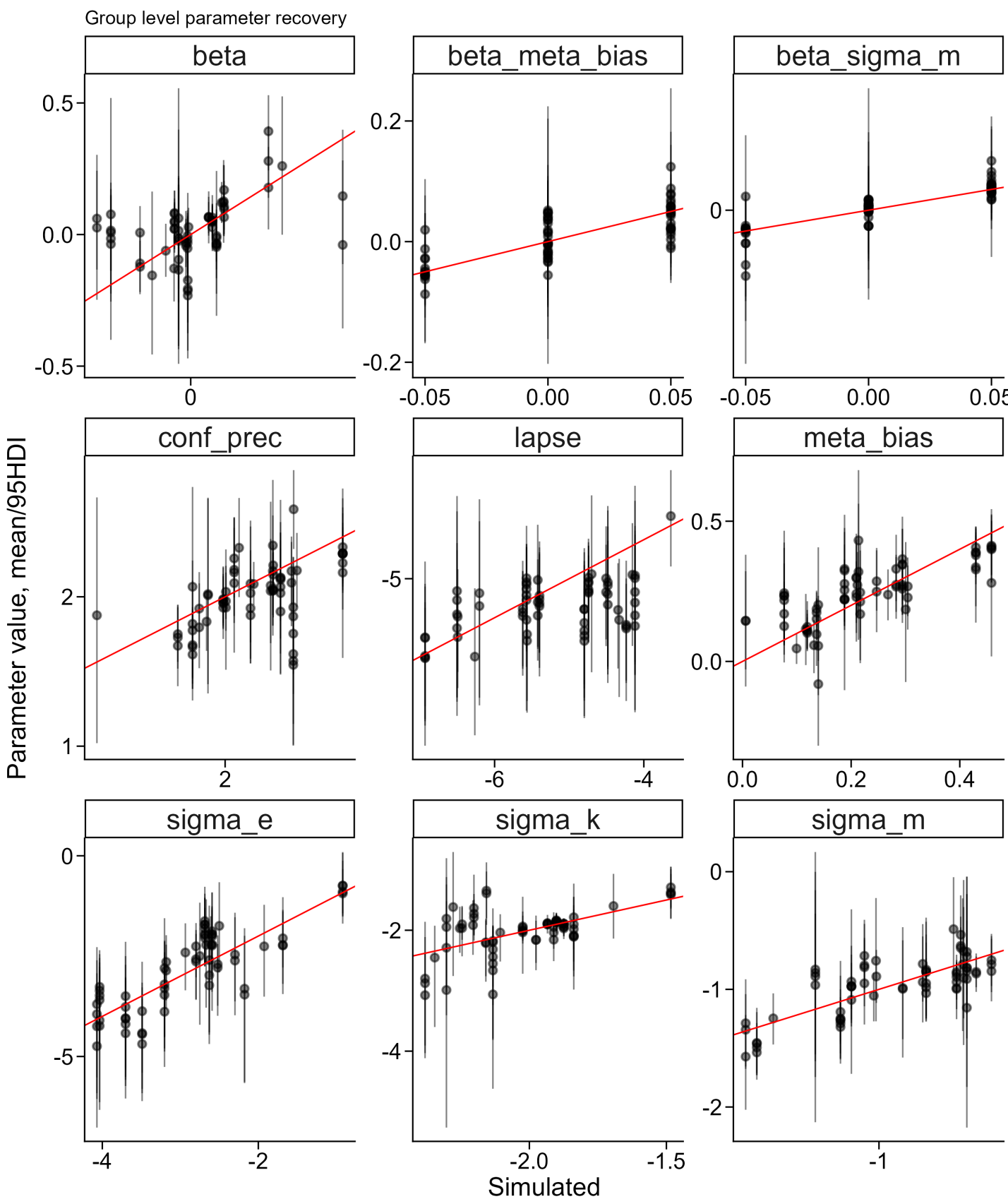
Posterior predictive checks

Next we show predictive checks of the Hierarchical heuristic model for our pilot participants.



Hierarchical parameter recovery

To further assess the robustness of our model implementation and the planned sequential-sampling procedure, we simulated new subjects from the posterior distribution of the hierarchical model. We did this by extracting a draw from the posterior and then simulating 25 subjects from this draw. We then fitted the model to increasing steps (i.e. 5,7,9, 12,15,20,25) of subjects, to demonstrate that the sequential sampling approach would be feasible. This also allowed us to do parameter recovery on the hierarchical model without additional computational resources. Below we showcase that the model can to an adequate degree recover its parameters on a subject and group level for parameters of interest.



Part 3: Design, Methods, and Hypotheses

Experimental study

The current study examines motion discrimination and meta-cognition in a Random Dot Motion (RDM) task. Specifically, we are interested in the question of whether post-decisional timing influences meta-cognitive uncertainty. We conceptualize this as whether confidence ratings follow the probability of being correct more closely when subjects are forced to wait longer before giving their confidence ratings, compared to when they can respond directly.

Dynamic models of meta-cognition assume that evidence is accumulated until a response boundary is reached and a binary decision is initialized. After the binary choice is given, additional evidence is assumed to be accumulated (i.e. post-decisionally) in order to give a confidence rating to the binary choice. If this post-decisional evidence accumulation is informative about the correctness of the choice, one would expect that increasing waiting time should increase the quality of the confidence ratings (i.e. confidence ratings would be closer to the objective probability that the choice was correct). However, if evidence used for the confidence rating is already computed around the time of the decision (i.e. pre-decisional) one would expect there to be no-difference or even expect the opposite effect. The reverse effect would indicate a situation where there exists some leakage of the information (perhaps due to memory constraints). In this study we manipulate the post decisional time window to examine its effects on the quality of the confidence ratings.

Previous studies investigating post decisional evidence accumulation have generally found that if post-decisional evidence accumulation exists it is easiest to detect in speed binary decision making tasks for easy trials, where increased post-decisional timing can decrease confidence ratings for incorrect choices. In this first study we set out to test the opposite case, a non-speeded perceptual decision making task with a range of trials from very easy to very hard (individually calibrated). As the evidence is not clear and we do not necessarily expect to find a significant difference in how post-decisional timing changes meta-cognitive uncertainty, we set out to test this in a Bayesian framework using sequential sampling. This not only let us provide evidence for the NULL-hypothesis (post-decisional time does not change the quality of the confidence ratings), but also allows us to collect as many participants as strictly needed to provide sufficient evidence.

Hypotheses

1. **Hypothesis 1:** Increasing the post-decisional time window will lead to no improvement or deterioration in the relationship between confidence and accuracy.

- **Test specification:**

This hypothesis is tested by allowing the σ_m parameter to vary by the trial-by-trial post-decisional time window. As this is the main parameter of interest, we will sequentially test whether this parameter has reached a Bayesfactor of either 30 or 1/30, which is our termination criteria.

2. **Hypothesis 2:** Increasing the post-decisional time window will lead to no difference in confidence bias.

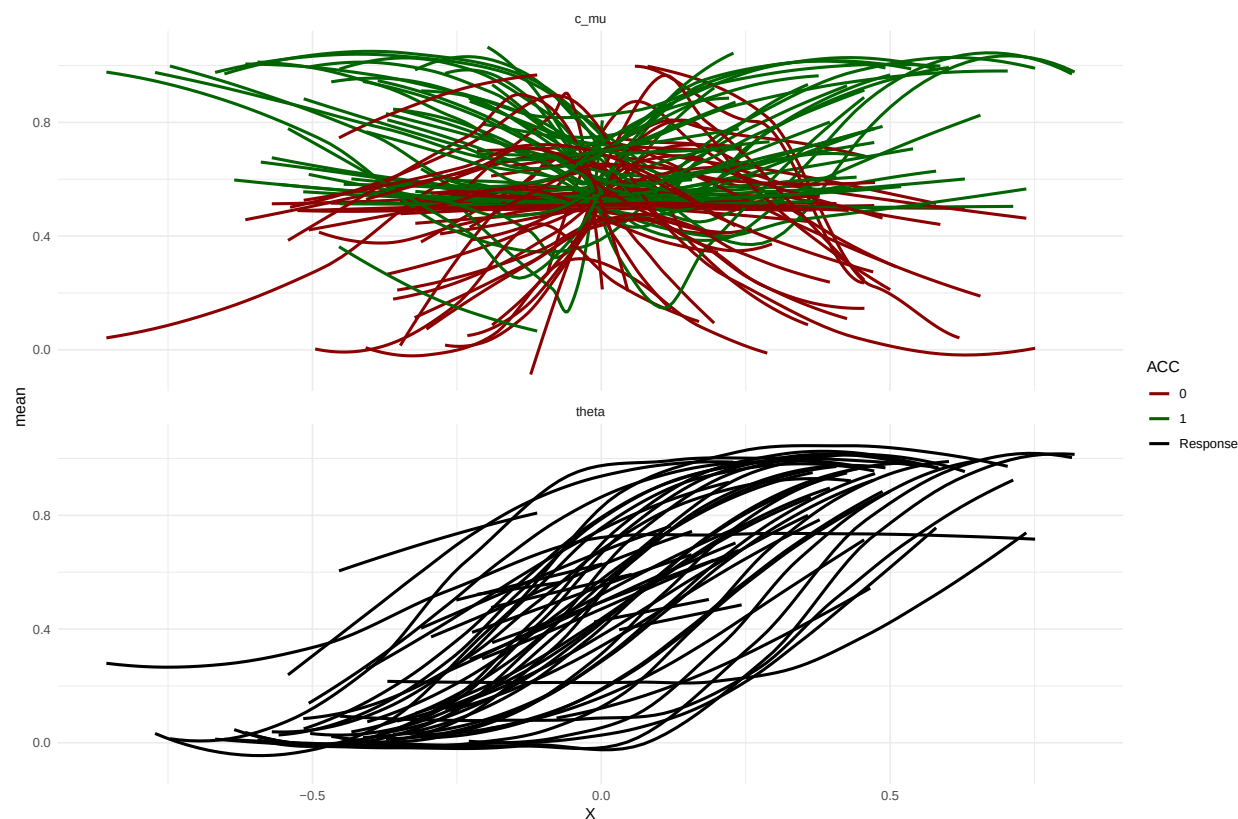
- **Test specification:**

This hypothesis is tested by allowing the μ_β parameter to vary by the post-decisional time window. The primary test evaluates whether the group-level parameter differs from 0 (i.e. does the 95 Highest density interval include 0), here we also aim to report the bayesfactor to determine the strength of the evidence compared to the NULL (i.e. there is no difference). This test will be conducted when the data collection has been terminated.

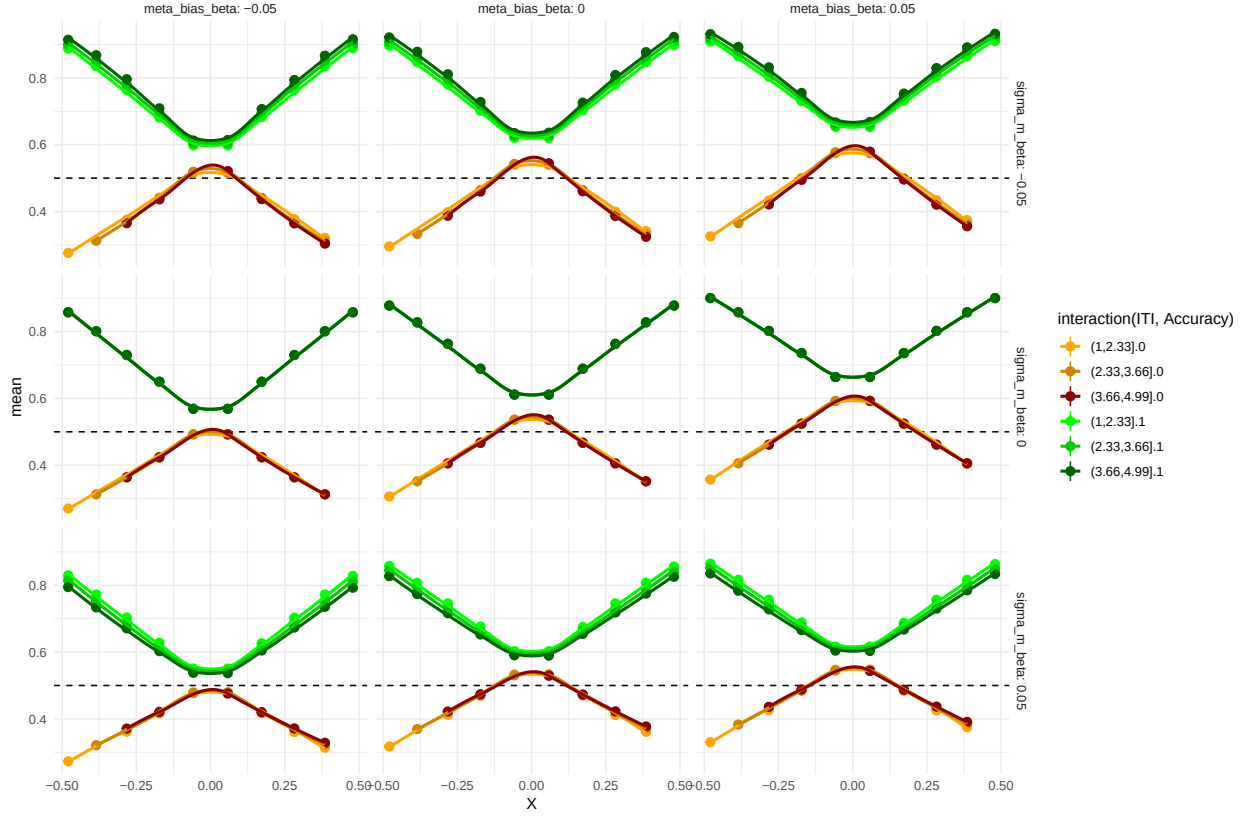
Priors and Sequential Sampling.

To limit the number of subjects needed to get sufficient evidence for our inference we will conduct a sequential sampling paradigm. Here we choose to terminate recruitment when the above explained Bayesfactor reaches either 30 or $1/30$, or we have collected 50 subjects. We will start examining the Bayesfactors in increments of 5 subjects when we have reached 15 total subjects. This is mainly due to stability of the model estimation, as well as expecting to test around 5 subjects per day.

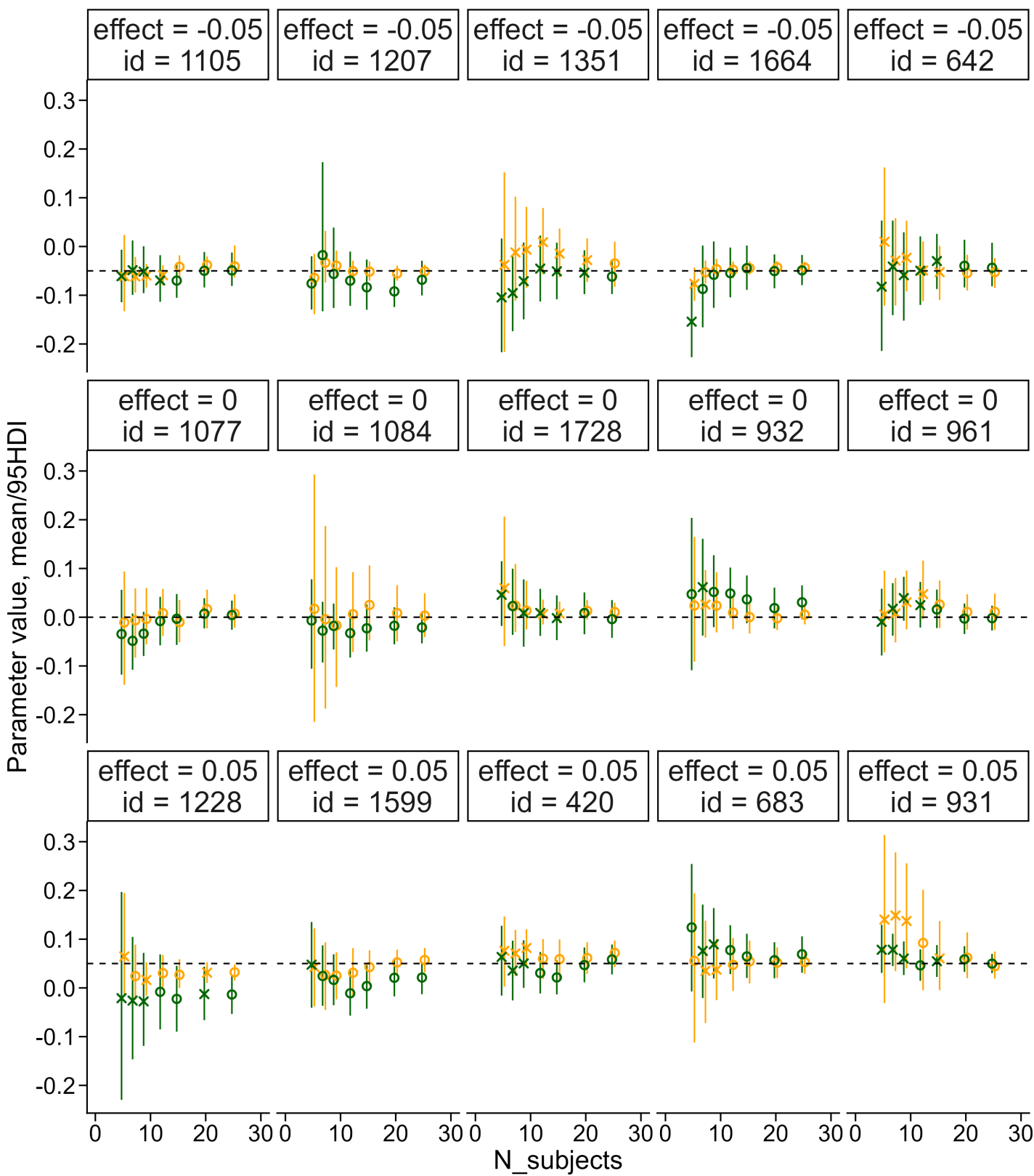
Priors As our sequential sampling approach uses bayesfactors for the termination of data collection, the prior distribution of this parameter is highly influential. To maximize transparency we visualize the curves that we allow given the prior below.



The plot above shows the prior predictive checks of the group level parameters (i.e. these curves represent group level curves). As can be seen the curves generally span everything. Now in order to test whether our sequential sampling approach works we simulate data from draws of the posterior distribution of our pilot data, we then investigate how the bayesfactors of both hypotheses outlined above evolve over time. Instead of drawing the parameters of interest we fix these to either -0.05, 0 or 0.05 to test that we can detect effects of this magnitude in both direction, but also terminate sampling if there is no effect. Below we display the size of these effects (simulated data from the mean of the group means).



Sequential sampling As mentioned above we simulated subjects from draws of the posterior that was informed by our pilot data. We then fitted the model to increasing steps (i.e. 5,7,9, 12,15,20,25) of subjects, to demonstrate that the sequential sampling approach would be feasible. below we showcase 5 such trajectories of Bayes factors across the number of subjects used, together with another panel showing how the posterior distribution of the parameter of interest changes with increasing subjects.



In the above plot we show the development of the effect of the inter-stimulus-interval on both meta cognitive bias (green) and meta cognitive uncertainty (orange). Circles display a fitted model without divergences above 400 effective sample size for both the bulk and the tails and rhat values below 1.03. Left side displays the updating of the posterior distribution of each parameter for increasing subjects. The right side displays the development of the Bayes-factors for the same draw of the posterior fitted to our pilot data.

As can be seen from the simulated sequential sampling we might run into a problem of getting bad diagnostics from the sampler even with this marginalized heuristic model. Here we firstly define our initial sampler parameters, then what we deem sufficiently good sampler diagnostics and lastly contingency plans if we do not get convergence.

Initial Sampler Configuration The model is initially estimated using the following sampler settings:

- Number of chains: 4
- Number of iterations per chain: 1000
- Warmup iterations: 1000
- Target acceptance rate (`adapt_delta`): 0.95
- Maximum tree depth (`max_treedepth`): 12

These values represent our baseline configuration for model estimation.

Convergence Criteria Sampler convergence and reliability will be assessed using the following diagnostics:

- $\hat{R}$ values below 1.03 for all group level parameters
- Effective sample size (ESS) above 400 for all group level parameters
- Zero divergent transitions after warmup
- No saturation of the maximum tree depth
- well-mixing trace-plots

If these criteria are met, the sampling procedure will be considered satisfactory.

Contingency Procedures If the above convergence criteria are not met, we will apply the following steps depending on the convergence failure:

1. Increase the number of iterations.
2. Increase the target acceptance rate (`adapt_delta`).
3. Increase the maximum tree depth (`max_treedepth`).
4. Reconsider model parameterization (i.e. centered vs non-centered parameterization).
5. Estimate the lapse-rate parameter non-hierarchically

These steps are intended to improve sampler stability and ensure reliable posterior estimation.

Interested readers are referred to the directory used to perform the [Sequential Sampling](#). Here the full Stan model and the scripts for checking diagnostics and model fit are present.

Participants

Eligible participants will be 18–40 years old, have normal or corrected-to-normal vision, and be proficient in English. Individuals with neurological or psychiatric diagnoses or those taking medication that may affect cognitive performance are also not eligible to participate. Subjects will be recruited through COBE Lab its participant pool to insure fast and effective recruitment and will be rewarded with 110DKK.

Design

This study consists of a single lab visit where participants perform a computer experiment on a COMPUTERSPEC(XXX) run using Psychopy (Peirce et al. 2019). The experiment uses a random dot motion (RDM) decision-making task. On each trial, participants view a random dot kinematogram in which a majority of dots move either upward or downward. While the dots are displayed, participants indicate whether the overall motion is ‘up’ or ‘down’ by pressing a corresponding key. They have 2 seconds to respond, trials without a response within this window are classified as no-response trials. After indicating their response, participants view a post-decisional fixation cross. The duration of this interval is the main experimental manipulation and is randomly drawn from a uniform distribution ranging from the maximum response time of 2 seconds to 5 seconds ($PostD_i \sim U(2, 5)$). After the interval, participants provide a metacognitive evaluation of their decision. They are asked, ‘How confident were you in your decision?’ and respond using a scale ranging from ‘certainly wrong’ to ‘certainly right.’ Participants navigate the scale using the arrow keys and confirm their response with a key press. If no response is made within 3 seconds, the trial is classified as a no-response trial. Before the start of each new trial, an inter-trial interval is presented consisting of a fixation cross. The duration of this interval is drawn from a log-normal distribution with a mean of 1 seconds and a standard deviation of 0.1 ($ITI_i \sim \text{Lognormal}(1, 0.1)$).

In total, participants will complete 288 trials, spread over 8 blocks of 36 trials. Before the main experiment begins, participants receive instructions on how to perform the decision task. They then complete six easy, non-speeded practice trials without confidence ratings, followed by 120 more difficult practice trials. Afterward, participants receive additional instructions about the confidence scale and complete 10 easy, non-speeded practice trials that include confidence ratings, after which the main experiment begins. The key variables of interest are response choice, accuracy, response times, confidence ratings, and the response times associated with those confidence ratings.

To ensure that task difficulty is calibrated at the individual level, the 120 harder practice trials are used as part of a staircase procedure. In a random dot kinematogram, task difficulty is determined by several stimulus properties, which can be broadly grouped into two categories: stimulus parameters and stimulus noisiness. To ensure that difficulty is optimally calibrated for each participant, we use the first 40 of the harder practice trials to staircase the dotlife parameter in the Dotstimclass in [psychopy](#). This is a stimulus parameter that determines how many frames a dot lives before disappearing. Once an appropriate dot-life value is identified, we start staircasing coherence. Coherence controls the stimulus signal-to-noise ratio and is the most commonly manipulated difficulty parameter in the literature. It ranges from 0 to 1 and determines the proportion of signal to noise dots, with higher values indicating a greater percentage of signal dots.

The remaining 80 trials are used to estimate the slope and threshold of the participant’s psychometric function on coherence. Both staircases are implemented in PsychoPy using the QUEST+ Bayesian adaptive procedure, which selects each stimulus to minimize posterior entropy over the psychometric parameters (Watson, 2017; implementation: <https://www.psychopy.org/api/data.html#psychopy.data.QuestPlusHandler>). Using the estimated psychometric function, we determine the coherence levels for the main experiment based on a set of predefined target accuracies ($ACC = [0.51, 0.6, 0.7, 0.8, 0.99]$). The corresponding coherence values are then sampled in a quasi-normal fashion: the extreme accuracy levels (0.51 and 0.99) appear once per direction, the intermediate levels appear twice, and the 0.70 level appears three times. In each block of the main experiment, trials are presented in a pseudo-randomized sequence of upward and downward motion such that no more than three consecutive trials share the same direction. This reduces the risk of implicit directional bias. All of this ensures that participants use the full confidence scale and that difficulty is balanced across upward and downward motion directions.

Exclusion criteria

Because the task is a two-alternative forced-choice decision, chance performance is 50%. For each participant, we will compute overall accuracy across all main-task trials and will exclude all participants with an accuracy lower than 60%. For our trial count, accuracy below this threshold is not reliably above chance under a binomial model of guessing. Furthermore, given the stimulus calibration procedure, an optimal observer is expected to achieve approximately 71% accuracy.

Taken together, an accuracy below 60% is inconsistent with meaningful participation in the task. We will also exclude participants who have a no-response rate greater than 30% (combining decision and confidence responses), as a high proportion of missed responses also indicates insufficient engagement with the task.

Physiological measures

As an additional stream of data we also collect physiological measures from each of our subjects. Specifically, we will be collecting eyetracking (and pupil size) using an Eyelink 1000, furthermore we will collect electrocardiogram (ECG) and respiration measures using Biopac Systems. To ensure calibration of the eyetracker the experimental script re calibrates after each experimental block, which also allows for frequent breaks for the participant.

References

- Baranski, J. V., and W. M. Petrusic. 1998. "Probing the Locus of Confidence Judgments: Experiments on the Time to Determine Confidence." *Journal of Experimental Psychology. Human Perception and Performance* 24 (3): 929–45. <https://doi.org/10.1037//0096-1523.24.3.929>.
- Baranski, J. V., and W. M. Petrusic. 2001. "Testing Architectures of the Decision-Confidence Relation." *Canadian Journal of Experimental Psychology = Revue Canadienne De Psychologie Experimentale* 55 (3): 195–206. <https://doi.org/10.1037/h0087366>.
- Caziot, Baptiste, and Pascal Mamassian. 2021. "Perceptual Confidence Judgments Reflect Self-Consistency." *Journal of Vision* 21 (12): 8. <https://doi.org/10.1167/jov.21.12.8>.
- Desender, Kobe, K. Richard Ridderinkhof, and Peter R. Murphy. 2021. "Understanding Neural Signals of Post-Decisional Performance Monitoring: An Integrative Review." *eLife* 10 (August): e67556. <https://doi.org/10.7554/eLife.67556>.
- Fleming, Stephen M. 2017. "HMeta-d: Hierarchical Bayesian Estimation of Metacognitive Efficiency from Confidence Ratings." *Neuroscience of Consciousness* 2017 (1): nix007. <https://doi.org/10.1093/nc/nix007>.
- Fleming, Stephen M., and Nathaniel D. Daw. 2017. "Self-Evaluation of Decision-Making: A General Bayesian Framework for Metacognitive Computation." *Psychological Review* 124 (1): 91–114. <https://doi.org/10.1037/rev000045>.
- Fleming, Stephen M., Elisabeth J. van der Putten, and Nathaniel D. Daw. 2018. "Neural Mediators of Changes of Mind about Perceptual Decisions." *Nature Neuroscience* 21 (4): 617–24. <https://doi.org/10.1038/s41593-018-0104-6>.

- Fung, Herrick, Medha Shekhar, Kai Xue, Manuel Rausch, and Dobromir Rahnev. 2025. “Similarities and Differences in the Effects of Different Stimulus Manipulations on Accuracy and Confidence.” *Consciousness and Cognition* 136 (November): 103942. <https://doi.org/10.1016/j.concog.2025.103942>.
- Goueytes, Dorian, François Stockart, Alexis Robin, et al. 2025. “Evidence Accumulation in the Pre-Supplementary Motor Area and Insula Drives Confidence and Changes of Mind.” *Nature Communications* 16 (1): 6998. <https://doi.org/10.1038/s41467-025-61744-8>.
- Hangya, Balázs, Joshua I. Sanders, and Adam Kepecs. 2016. “A Mathematical Framework for Statistical Decision Confidence.” *Neural Computation* 28 (9): 1840–58. https://doi.org/10.1162/NECO_a_00864.
- Kiani, Roozbeh, and Michael N. Shadlen. 2009. “Representation of Confidence Associated with a Decision by Neurons in the Parietal Cortex.” *Science* 324 (5928): 759–64. <https://doi.org/10.1126/science.1169405>.
- Kubinec, Robert. 2023. “Ordered Beta Regression: A Parsimonious, Well-Fitting Model for Continuous Data with Lower and Upper Bounds.” *Political Analysis* 31 (4): 519–36. <https://doi.org/10.1017/pan.2022.20>.
- Maniscalco, Brian, and Hakwan Lau. 2012. “A Signal Detection Theoretic Approach for Estimating Metacognitive Sensitivity from Confidence Ratings.” *Consciousness and Cognition, Beyond the Comparator Model*, vol. 21 (1): 422–30. <https://doi.org/10.1016/j.concog.2011.09.021>.
- Mihali, Andra, Marianne Broeker, Florian D. M. Ragalmuto, and Guillermo Horga. 2023. “Introspective Inference Counteracts Perceptual Distortion.” *Nature Communications* 14 (1): 7826. <https://doi.org/10.1038/s41467-023-42813-2>.
- Navajas, Joaquin, Chandni Hindocha, Hebah Foda, Mehdi Keramati, Peter E. Latham, and Bahador Bahrami. 2017. “The Idiosyncratic Nature of Confidence.” *Nature Human Behaviour* 1 (11): 810–18. <https://doi.org/10.1038/s41562-017-0215-1>.
- Öztel, Tutku, and Fuat Balcı. 2024. “Metric Error Monitoring as a Component of Metacognitive Processing.” *European Journal of Neuroscience* 59 (5): 807–21. <https://doi.org/10.1111/ejn.16182>.
- Peirce, Jonathan, Jeremy R. Gray, Sol Simpson, et al. 2019. “PsychoPy2: Experiments in Behavior Made Easy.” *Behavior Research Methods* 51 (1): 195–203. <https://doi.org/10.3758/s13428-018-01193-y>.
- Petrusic, William M., and Joseph V. Baranski. 2003. “Judging Confidence Influences Decision Processing in Comparative Judgments.” *Psychonomic Bulletin & Review* 10 (1): 177–83. <https://doi.org/10.3758/BF03196482>.
- Pleskac, Timothy J., and Jerome R. Busemeyer. 2010. “Two-Stage Dynamic Signal Detection: A Theory of Choice, Decision Time, and Confidence.” *Psychological Review* (US) 117 (3): 864–901. <https://doi.org/10.1037/a0019737>.
- Rausch, Manuel, and Michael Zehetleitner. 2019. “The Folded X-pattern Is Not Necessarily a Statistical Signature of Decision Confidence.” *PLOS Computational Biology* 15 (10): e1007456. <https://doi.org/10.1371/journal.pcbi.1007456>.
- Resulaj, Arbora, Roozbeh Kiani, Daniel M. Wolpert, and Michael N. Shadlen. 2009. “Changes of Mind in Decision-Making.” *Nature* 461 (7261): 263–66. <https://doi.org/10.1038/nature08275>.
- Sánchez-Fuenzalida, Nicolás, Simon van Gaal, Stephen M. Fleming, Julia M. Haaf, and Johannes J. Fahrenfort. 2025. “Confidence Reports During Perceptual Decision Making Dissociate from Changes in Subjective Experience.” *Communications Psychology* 3 (1): 81. <https://doi.org/10.1038/s44271-025-00257-y>.

- Sanders, Joshua I., Balázs Hangya, and Adam Kepecs. 2016. “Signatures of a Statistical Computation in the Human Sense of Confidence.” *Neuron* 90 (3): 499–506. <https://doi.org/10.1016/j.neuron.2016.03.025>.
- Shekhar, Medha, and Dobromir Rahnev. 2021. “Sources of Metacognitive Inefficiency.” *Trends in Cognitive Sciences* 25 (1): 12–23. <https://doi.org/10.1016/j.tics.2020.10.007>.
- Tsetsos, Konstantinos, Thomas Pfeffer, Pia Jentgens, and Tobias H. Donner. 2015. “Action Planning and the Timescale of Evidence Accumulation.” *PLOS ONE* 10 (6): e0129473. <https://doi.org/10.1371/journal.pone.0129473>.
- Xue, Kai, Herrick Fung, and Dobromir Rahnev. 2026. “Stimulus Reliability but Not Boundary Distance Manipulations Violate the Folded-X Pattern of Confidence.” *Cognition* 272 (July): 106490. <https://doi.org/10.1016/j.cognition.2026.106490>.
- Yeung, Nick, and Christopher Summerfield. 2012. “Metacognition in Human Decision-Making: Confidence and Error Monitoring.” *Philosophical Transactions of the Royal Society B: Biological Sciences* 367 (1594): 1310–21. <https://doi.org/10.1098/rstb.2011.0416>.
- Yu, Shuli, Timothy J. Pleskac, and Matthew D. Zeigenfuse. 2015. “Dynamics of Postdecisional Processing of Confidence.” *Journal of Experimental Psychology. General* 144 (2): 489–510. <https://doi.org/10.1037/xge0000062>.